Trading Volume Alpha*

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Abstract

We model the economic consequences of predicting individual stock trading volume by translating volume forecasts into a component of expected trading costs, and analyzing their value via portfolio optimization that trades off tracking error versus net-of-cost performance. Incorporating economic losses directly into the prediction problem yields substantial improvements in out-of-sample performance. Economic losses, unlike statistical losses, are asymmetric and vary with fund size. Applying these insights with machine learning to real data, we find that volume prediction can be as valuable as return prediction for portfolio decisions, highlighting the potential role for predicting non-price-based variables for asset pricing.

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1 Introduction

Research in asset pricing primarily focuses on predicting moments of asset returns.[1] However, other non-return forecastable variables may be valuable for asset pricing and investment decisions. We focus on one such variable – asset-specific trading volume. Although not the first to look at predicting trading volume,[2] we take a unique approach linking it directly to asset pricing and portfolio choice. We examine the economic consequences of predicting volume by recasting the volume prediction problem into a portfolio problem. Our model evaluates the cost of trading versus the (opportunity) cost of *not* trading (i.e., tracking error) to maximize net-of-cost performance, where trading costs and tracking error are endogenously negatively related.

We connect predicted trading volume to expected trading costs using a simple version of Kyle's (1985) model, where price impact decreases with trading volume. Holding returns fixed, the improvement we find in net-of-cost alpha from volume prediction alone, we term "trading volume alpha." All else equal, a higher predicted volume allows the agent to trade more aggressively (i.e., reduce tracking error) because price impact per dollar traded is expected to be lower. Reframing the optimization as a tracking error problem serves several purposes. First, it avoids the well-trodden ground and usual pitfalls of return prediction.[3] Second, it makes clear the economic impact of volume prediction separate from return prediction to avoid optimizing dual objectives. Finally, this framework can apply broadly to settings with no return/alpha prediction, such as hedging, benchmarking to an index, satisfying an investment constraint (e.g., ESG), or executing an arbitrage.

Quantifying the economic benefits of non-return variable prediction is challenging and requires connecting, through theory or empirical work, the variable to an economic outcome. Our translation of volume into expected trading costs is purposely simple. Our goal is not to provide the best or most reliable trading cost model or forecast, but rather to simply translate volume prediction into

[1] A long literature in asset pricing focuses on mean return prediction, with summaries, including some criticisms, found in Harvey, Liu, and Zhu (2016); McLean and Pontiff (2016); Jensen, Kelly, and Pedersen (2023). A good summary of the literature on return volatility prediction is Engle (2004).

[2] See literature review in Section 2.

[3] The literature shows return predictability decays over time and with publication (McLean and Pontiff, 2016; Linnainmaa and Roberts, 2018; Chen and Zimmermann, 2020), and there is much debate that the search for return prediction has been oversaturated (Harvey, Liu, and Zhu, 2016; Jensen, Kelly, and Pedersen, 2023).

an economic primitive. Our aim is to provide a general forecast of costs for trading an individual stock in order to quantify the value of predicting a stock's volume. While trading costs have received some attention in the literature, *forecasting* costs has not. For this task, we purposely choose a simple, and generic, model.[4] In spite of this, we find that volume prediction has significant economic value in a variety of settings. We discuss where our model over- or understates the value of volume prediction.

More interestingly, converting volume prediction into economic costs changes the forecasting problem in a way that improves decision making. For example, an interesting feature that emerges from the model is that the economic cost of volume forecast errors is non-linear and asymmetric. Price impact costs are linear in participation rate, but non-linear in trading volume, and the cost of forecast errors is asymmetric, and varies across the size of the trade and the size of the manager/fund. As volume tends to zero, price impact costs are unbounded, whereas when volume is large, impact costs are negligible. This creates asymmetric costs of volume forecast errors that vary with different volume levels – overestimating volume when volume is low is particularly costly. Conversely, tracking error, or the opportunity cost of not trading, is (assumed) independent of volume and finite. The combination of these two effects means the optimization penalizes overestimating volume more than underestimating volume over the majority of the actual volume distribution. Intuitively, an illiquid security's price impact is very sensitive to small volume forecast errors, and in a highly non-linear and asymmetric way, whereas a highly liquid security's price is fairly inelastic to forecast errors.[5]

[4]Predicting trading costs, particularly at the individual stock level, is both complicated and challenging. The biggest component of costs for a large investor is price impact, which depends on the size of the trade, the amount traded by other traders in that security, and the urgency and identity of the trader (Frazzini, Israel, and Moskowitz, 2018). While many different trading cost models exist, they universally contain these three elements: trade size, market size, and trader specifics (identity, information, motive, patience, etc.). Theoretical foundations of trading costs in the classic market microstructure literature include asymmetric information (Kyle, 1985; Glosten and Milgrom, 1985) and inventory costs (Stoll, 1978; Ho and Stoll, 1983; Grossman and Miller, 1988), where volume is the major determinant of liquidity (Benston and Hagerman, 1974; Glosten and Harris, 1988; Brennan and Subrahmanyam, 1995; Campbell, Grossman, and Wang, 1993; Datar, Naik, and Radcliffe, 1998; Amihud, 2002). Our simple model circumvents these issues by focusing exclusively on the component of costs that is neither trader-specific nor endogenous – making it a generic expected trading cost model that uses only widely available volume data without having to find trader-specific and limited-access trading cost data (Frazzini, Israel, and Moskowitz, 2018).

[5]Our framework, and its implication that portfolio optimization will seek to trade conservatively rather than aggressively because of the asymmetric cost of volume forecast errors, implies that arbitrage activity may be limited as a consequence, especially in illiquid securities. This implication provides a novel and additional source of limits to arbitrage activity in the spirit of Shleifer and Vishny (1997).

It is not only trading volume that matters, but also the size of the trade relative to that volume (the "participation rate"). Trade size is determined endogenously and is a function of expected trading volume and aversion to tracking error. In the model, the size (assets under management, AUM) and volatility of the fund (and risk aversion of the investor) affect the costs and benefits of trading. Because price impact is an increasing function of participation rate, trading costs increase with AUM endogenously, and the relative penalty for tracking error decreases with AUM. The optimal tradeoff between trading costs and tracking error will therefore vary with the size of the portfolio, and so will the economic impact of volume prediction. For small AUM, tracking error considerations likely dominate trading cost considerations, hence the economic benefit to predicting volume is relatively less valuable. For large AUM, trading cost considerations dominate.

Incorporating these insights into the forecasting problem greatly improves the value of volume predictability over statistical loss functions, like MSE, that assume symmetry in forecast errors and are invariant to portfolio size. This result highlights the tension between statistical precision and economic decision-making, where forecast accuracy may be sacrificed in lieu of better economic outcomes. In this sense, our results relate to Leitch and Tanner (1991) and Brandt (1999), but where we uniquely apply this to trading volume and via machine learning.

Applying the model to data, we show that volume is highly predictable using a host of variables: technical signals, such as lagged returns and lagged trading volume, indicators for various market-wide and firm-level events, such as triple-witching and earnings releases, and firm characteristics tied to return anomalies. We use several linear and non-linear prediction methods, including neural and recurrent neural networks. We find large datasets of predictors and complex models provide the best forecast accuracy.

To assess the economic value of these predictions, we conduct several empirical exercises. First, as a simple test that requires no trading cost model assumptions, we forecast daily VWAP (volume-weighted average price), which represents the best traded prices one could have traded on that day. We ignore trying to predict prices, taking tomorrow's closing price as given, and simply use our volume prediction to forecast VWAP. The out-of-sample relative absolute error of our VWAP forecasts is as low as 0.43% over a month (value-weighted across stocks using the recurrent neural

net and all forecast variables) and as large as 1.33% using only a five-day moving average of volume and equal-weighted across stocks.

Second, we run a series of trading experiments that account for trading costs (with stock-level volume prediction as the sole input) using our portfolio model. With liquidity being an unknown quantity, the portfolio manager uses the volume forecast to endogenously respond to her expected cost and benefit of trading. The out-of-sample performance of this portfolio choice is then evaluated net of trading costs. More accurate volume predictions provide more efficient implementation and hence result in more efficient out-of-sample net-of-cost portfolio performance. We run these trading experiments in two ways. The first approach predicts volume to minimize a statistical error criterion and then inputs the statistically efficient forecast into the portfolio problem. This is the "classic" approach of forming statistically efficient parameter estimates in one stage and inputting them into an economic model in the second stage. The second approach combines everything into a single stage, where we inform the prediction problem about the economic objective. Specifically, we alter the objective/loss function to take into account the portfolio problem's economic objective when predicting volume. This second approach may sacrifice statistical accuracy for economic value, but may lead to better portfolio decisions. The idea relates to Brandt (1999), but we uniquely employ it within a machine learning framework via "transfer learning," and apply it to trading costs.

We start with a simulation of a highly profitable, but high turnover, trading strategy that delivers a huge before-cost profit but would not survive trading costs, if trading volume is ignored. This hypothetical portfolio is designed to mimic a profitable daily trading strategy that quantitative hedge funds would employ, but where attention to liquidity and trading costs is imperative to make it profitable. Ignoring volume and trading uniformly guarantees trading costs will wipe out investment capital. However, by predicting volume, the optimizer alters the frequency of trading, which stocks to trade, and the trade size (while balancing tracking error) to deliver positive net-of-cost returns. Varying the size of the fund leads to different tradeoffs, where larger funds rely more heavily on volume forecasts and make sharper tradeoffs due to bigger price impact.

The simulated hypothetical trading experiment is meant to illustrate how our model works with real volume data, where the economic value of volume prediction is large. While this example

is not different from the type of quantitative trading strategies many hedge funds deploy, to get a broader sense of how much volume prediction can matter for the portfolios the asset pricing literature focuses on, we also target the 153 factor portfolios of Jensen, Kelly, and Pedersen (2023). Here, we also find that volume prediction has a measurable and economically significant effect on after-cost portfolio performance. Volume prediction benefits all traded factors, but some factors have greater "trading volume alpha" than others, due to the varying tradeoff between the cost of trading and the opportunity cost of not trading across factors. Intuitively, factors with higher turnover and more sharply decaying alpha (e.g., short-term reversals, momentum) benefit the most from better volume forecasts. Factors with low turnover and little alpha decay benefit the least (e.g., profitability). And, these benefits vary with the size of the fund. For a $10 billion fund, the after-cost improvement in portfolio performance due solely to trading volume prediction (beyond just using lagged 5-day moving averages of volume), can double the net-of-cost expected returns and Sharpe ratio of some factors. The improvement in after-cost returns ranges from 23 bps (size factor) to 94 bps (short-term reversal factor) per year relative to just using a moving average of lagged volume to predict volume. These improvements are much larger if volume is ignored entirely, where some factors fail to cover trading costs at all if volume is ignored.

We conclude that "trading volume alpha" can be substantial, and potentially as large as return prediction alpha that dominates the literature. Refining the prediction methods and deepening the prediction models could add substantially to these improvements. Modeling other variables that predict trading costs in addition to volume, such as volatility, bid-ask spreads, and other microstructure variables, in a richer trading cost framework, could yield even larger benefits. Our aim is to simply show the power of forecasting a single non-return variable for asset pricing and portfolio choice. Despite its simplicity, our framework illustrates the importance of volume prediction models, which we hope sparks future studies to integrate market microstructure frictions into asset pricing forecasting tools (Goldstein, Spatt, and Ye, 2021).

More generally, the idea of forecasting non-return variables and translating them into economic inputs, may be a promising area for asset pricing research. Our insights that incorporating the economic translation into the prediction problem's objective yields better decision-making, while

sacrificing forecast accuracy, are very useful and have broader implications. A natural, if not challenging, next step is to combine multiple objectives of predicting volume, return moments, and other non-return variables simultaneously (and their potential interactions) by translating each into a common economic objective. We leave this to future research.

The rest of the paper is organized as follows. Section 2 presents some preliminaries for the analysis, including motivation, literature review, and data description. Section 3 examines volume prediction from a statistical perspective. Section 4 presents a theoretical framework for quantifying the economic value of volume prediction and Section 5 describes our empirical implementation of that framework through machine learning. Section 6 applies these insights and methods to real-world trading experiments. Section 7 concludes.

2 Preliminaries: Motivation, Literature Review, and Data

We motivate interest in predicting volume, review the literature, and describe our data.

2.1 Motivation

Motivating the prediction of non-return variables for asset pricing requires converting those variables into economic outcomes. For trading volume, a natural conversion is into some cost of liquidity or trading cost. Real-world portfolios face execution costs, which are critical to realizing investment performance. However, empirical challenges to modeling trading costs and finding available and generalizable data limit the attention it has received. One of the main challenges in modeling and forecasting trading costs is that the largest component of these costs for a large investor is price impact, which depends on the size of the trade, the amount traded by other traders (in the same and in opposite directions), and the identity of the trader. Different traders face different price impact. These features frustrate a generic portfolio solution that incorporates trading costs.[6]

[6]One solution is to use a universal benchmark for costs, like VWAP, which represents the average price traded across the volume of trades and is the best execution price available. However, VWAP can only provide a generic benchmark, it cannot be altered by the trader, and therefore offers no insight on price impact, which significantly limits its use as a decision-making tool. We use VWAP as a benchmark comparison to understand the economic value of volume forecasts, but provide a richer analysis of the value of volume forecasts using a price impact model.

Following Kyle (1985) and subsequent empirical work (Frazzini, Israel, and Moskowitz, 2018), price impact depends on the trader's participation rate, defined as the dollar amount traded by an investor in a security relative to total dollar volume traded in that security at the same time. Price impact is an increasing function of participation rate, whose elasticity varies by investor, where the amount traded by each investor is endogenous to expected price impact, which itself is a function of the participation rate. The circular nature of trade size, and its variation across investors, makes modeling price impact particularly challenging. However, the denominator of the participation rate is exogenous to the trader's desired trade size (assuming a single investor's trade is too small to materially affect total volume), and is universal to all investors. Moreover, from an empirical standpoint, total dollar volume is readily available at high frequency, while data on individual traders is not. We therefore model expected trading costs using only forecasted total dollar volume for a stock in order to translate volume into an economic input. This simple model is only a partial solution to the trading cost problem, but it is a general one, and, more importantly, one that allows us to showcase the economic value of predicting trading volume. The goal is not to produce the best trading cost model, but rather to translate a non-return variable (volume) into an economic object for portfolio implementation and then to see how valuable that variable is.

2.2 Literature Review

Although not the first to examine predicting trading volume,[7] our examination is unique in several respects. First, we tie volume prediction directly to an economic outcome – expected trading costs – to quantify its value in a portfolio context. This provides a tighter link between volume prediction and asset pricing. Lo and Wang (2000) argue that trading volume has received far less attention in asset pricing research compared to prices and returns, advocating for greater research attention to volume. Two and a half decades later, volume research still receives relatively little attention, in part perhaps, because of the challenge in linking it to economic consequences. Second, we forecast

[7] Gallant, Rossi, and Tauchen (1992); Brock and LeBaron (1996); Darrat, Rahman, and Zhong (2003) study volume dynamics and predictability, focusing on lead-lag relationships between volume and return volatility for the S&P 500 stock index and individual stocks. Chordia, Huh, and Subrahmanyam (2007) and Chordia, Roll, and Subrahmanyam (2011) examine the determinants of trading volume activity, using proxies for firms' visibility, uncertainty, and dispersion of opinion.

volume using machine learning techniques and a rich set of predictors.[8] Third, by translating volume prediction into an economic input and incorporating it into the optimization, we alter the forecasting objective because economic losses are not the same as statistical losses, resulting in *less accurate* volume prediction, but *more valuable* volume forecasts for portfolio decisions. This result is reminiscent of Brandt (1999), but applied in a different context and with different methods.

Our work also contributes to research on transaction costs and portfolio implementation.[9] The key difference between these papers and ours is that these previous papers take transaction costs as given and known to the investor. In our setting, trading costs are unknown and must be forecasted and the error in those forecasts plays an important role in portfolio optimization, something previous studies have ignored.

To better understand our contribution and how it differs from the literature, consider the most recent paper by Jensen et al. (2024), who evaluate the performance of ML models (for return prediction and for portfolio choice) given assumed and known transaction costs. Taking transaction costs as given, they ask whether ML models that predict returns are still useful after adjusting for known transactions costs and whether ML models can be trained to optimize in light of these given costs.[10] Their objective is about after-cost evaluation of signals and models for given trading costs, which is very different from our objective, which is to evaluate the economic content of non-return predictors, in this case trading volume, and where trading costs are unknown. The fact that we use an expected trading cost model to give trading volume forecasts economic content connects these papers, but fundamentally our aim is different. Our paper studies a related problem, but comes at it from the opposite direction. We emphasize that transaction costs are *not* given or known when forming the portfolio, and that the investor must make her allocation decisions based on *expected* transactions costs, which are estimated with error. We ask what is the economic value

[8] Kaastra and Boyd (1995) is an early precursor that applies neural networks as a prototypical artificial intelligence method to forecast futures volume for the Winnipeg Commodity Exchange (WCE).

[9] Korajczyk and Sadka (2004); Frazzini, Israel, and Moskowitz (2012); DeMiguel et al. (2020); Chen and Velikov (2023); Detzel, Novy-Marx, and Velikov (2023); Avramov, Cheng, and Metzker (2023); Azevedo, Hoegner, and Velikov (2024); Jensen et al. (2024) examine the performance of anomaly portfolios and (or) machine learning-driven portfolios after accounting for transaction costs.

[10] This work follows Brandt, Santa-Clara, and Valkanov (2009) who directly model portfolio weights and optimize after-cost portfolio performance. Simon, Weibels, and Zimmermann (2025) upgrade their framework to deep neural networks.

of having more accurate trading cost forecasts using a portfolio framework? And, what is the cost of trading cost forecast error? By translating volume into expected trading costs, we summarize the value of volume forecast improvement in terms of portfolio economics.[11] This translation allows us to evaluate forecast improvements in terms of meaningful economic consequences rather than statistical metrics such as R^2 or MSE. In addition, we find that incorporating that economic objective into the forecasting problem directly yields significantly better OOS results, offering insight into ways to improve forecasting models, including from machine learning, by merging economic modeling with computer science techniques.[12] In this sense, our results relate to Leitch and Tanner (1991) and Brandt (1999), but updated to incorporate new ML techniques and applied in a novel setting.

2.3 Data

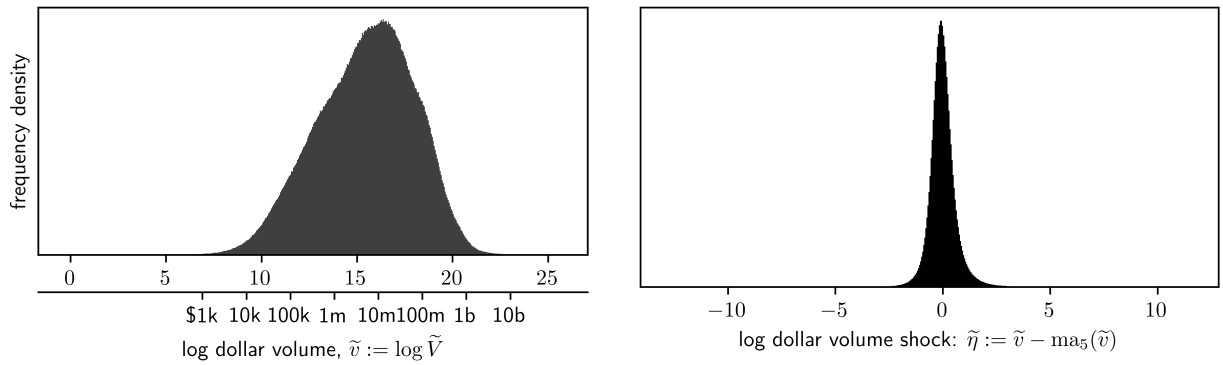
We compile a data panel of daily stock-level dollar trading volume ($\widetilde{V}_{i,t}$) and 175 predictors ($X_{i,t}$). The unit of observation is stock-day (i,t). We adopt the convention that $X_{i,t}$ is observed by day $t-1$, whereas the associated prediction target $\widetilde{V}_{i,t}$ is observed until the end of day t. A tilde denotes a random variable conditional on the information prior to day t.

The sample period is 2018 to 2022 (1,258 days). The cross-section covers around 4,700 U.S. stocks, with an average of 3,500 stocks per day (4.4 million observations). We split the data into a 3-year training sample and a 2-year testing sample. All models are trained once in the training sample and evaluated out of sample. The short, but recent, time period is chosen because trading volume and especially trading costs have changed dramatically over time, where costs have come down considerably over time, including price impact (Frazzini, Israel, and Moskowitz, 2018). The short sample therefore likely understates the economic importance of our results, yet still provides

[11] Related to this idea, Białkowski, Darolles, and Le Fol (2008) model intraday volume dynamics to improve daily VWAP execution. Balduzzi and Lynch (1999) and Çetin, Jarrow, and Protter (2004) model transaction costs in portfolio settings and study the economic value from various transaction cost models.

[12] From a methodology standpoint, our application of transfer learning is closely related to Chen et al. (2023). We present an "economic learning" method that fine-tunes statistical forecasts on an economic model-implied objective in order to optimize economic value. Chen et al. (2023) apply transfer learning from a source domain of simulated data to "teach economics" to the model. In our case, the transfer is not between different datasets, but between different objective functions. Other recent machine learning applications in portfolio optimization also use the portfolio outcome as the direct objective function, for example Chen, Pelger, and Zhu (2024), Cong et al. (2020), following the seminal paper of Brandt, Santa-Clara, and Valkanov (2009).

Figure 1: Distributions of daily stock-level log dollar volume ($\widetilde{v}$) and its shock ($\widetilde{\eta}$), panel pooled



Histograms of $\widetilde{v}_{i,t}$ and $\widetilde{\eta}_{i,t}$ in the full sample of around 4,400,000 stock-day observations. The second horizontal axis in the left panel is dollar volume $\widetilde{V}_{i,t}$ in the log scale. Log dollar volume shock $\widetilde{\eta}$ is daily log dollar volume minus the moving average in the past five days: $\widetilde{\eta}_{i,t} := \widetilde{v}_{i,t} - \frac{1}{5}\left(\widetilde{v}_{i,t-1} + \cdots + \widetilde{v}_{i,t-5}\right)$.

a large enough sample for statistical precision.

Our analysis focuses on predicting out-of-sample daily trading volume. Reasonable in-sample fit, often evaluated in the literature (Chordia, Huh, and Subrahmanyam, 2007; Chordia, Roll, and Subrahmanyam, 2011), does not often lead to good out-of-sample (OOS) performance, which is of primary interest to evaluate the robustness and economic impact of volume predictability.

2.4 Prediction objects: daily stock trading volume

The main variable we aim to predict is daily dollar trading volume, which we measure as the natural logarithm of end-of-day transacted total dollar trading volume for each stock. Daily dollar trading volume $\widetilde{V}_{i,t}$ ranges widely (from thousands to billions of dollars) across stocks and is highly skewed. We take the log of dollar volume, $\widetilde{v} = \log \widetilde{V}$, whose distribution, shown in Figure 1, is relatively well-behaved, being close to normal and symmetrical. Log dollar volume is highly persistent, and can be easily predicted by lagged moving averages of various frequencies. The five-day moving average predicts log dollar volume with an R^2 of 93.68%, higher than the one-day lag (92.53%), moving average of 22 days (92.60%), or 252 days (86.12%).

We focus on predicting the log dollar volume *shock* defined as daily log dollar volume minus the moving average over the previous five days, $\widetilde{\eta}_{i,t} := \widetilde{v}_{i,t} - \frac{1}{5}\left(\widetilde{v}_{i,t-1} + \cdots + \widetilde{v}_{i,t-5}\right) := \widetilde{v}_{i,t} - [\mathrm{ma}_5]_{i,t}$. Predicting the shock is equivalent to predicting the level as one can always add the baseline back to

the shock for the level forecast. Taking out the baseline is comparable to predicting asset returns (change in log price) instead of price levels. We select the five-day frequency due to its highest R^2. Figure 1 (right panel) shows the pooled distribution of $\widetilde{\eta}$, which is relatively symmetric, centered around zero, and has long tails.

2.5 Predictors

While it is common in the stock return predictability literature to use large sets of variables (e.g., "factor zoo") to identify the best predictors, it is less common to use large data sets to predict market microstructure variables such as trading volume (Easley et al., 2020). We show that using big data improves the precision of volume forecasts significantly. We use a total of 175 predictors from various sources, including technical signals, firm fundamentals, and market and corporate events. We show that the virtue of complexity approach in return prediction (Kelly, Malamud, and Zhou, 2024) is also useful for volume prediction. Each subset of variables provides incremental improvement to predicting volume, while using all variables has the greatest out of sample predictability. The sets of predictors that are cumulatively added to the prediction model are:

1. Technical signals ("tech"): lagged moving averages of returns and log dollar volume over the past 1, 5, 22, and 252 days. (8 predictors.)
2. A small set of commonly used fundamental firm characteristics ("fund-1"): market equity, standardized earnings surprise, book leverage, book-to-market equity, Dimson beta, and firm age. (6 predictors.)[13]
3. The remaining firm characteristics from the JKP dataset ("fund-2"), which are merged and transformed in the same way as fund-1 variables. (147 predictors.)
4. Calendar dates that may affect trading volume ("calendar"): early closing days for the exchanges (July 3rd, Black Friday, Christmas Eve, and New Year's Eve); triple witching days (four times a year when the index futures contract expires at the same time as the index option contract and single stock options); double witching days (eight times a year when two

[13]These predictors are from the JKP dataset (Jensen, Kelly, and Pedersen, 2023). We forward fill the monthly firm characteristics in time when merging to the daily panel. Hence, the characteristics are still always ex-ante available. On each day, we rank standardize in the cross-section each characteristic to a uniform distribution from -1 to +1.

of the three events above, the single stock options and index options expiration, coincide); Russell index re-balancing (once a year, the fourth Friday in June).[14]. Early closing days have substantially less trading volume, while the other three are associated with positive spikes in trading volume.

5. Earnings release schedule ("earnings"): 10 categorical dummy variables indicating whether the firm has an upcoming earnings release or just had one in the past few days. We first construct the number of days until the next known scheduled earnings release event. For example, a value of zero implies the current day is previously known to have a scheduled release. A negative value means there are no known scheduled events in the future and indicates how many days since the last event. We convert this number into 10 dummy variables of categorical bins: $\leq -4, -3, -2, -1, 0, 1, 2, 3, 4, \geq 5$. The data source is the Capital IQ Key Developments dataset. (10 predictors.)

This list of variables for predicting volume is not exhaustive. Other variables that could add predictive power are microstructure variables, intraday observations, and lead-lag relations across stocks in terms of trading (e.g., large to small stocks, within industry, etc.). As stated previously, we do not attempt to provide the best volume prediction model. Rather, we translate the volume prediction problem into an economic problem whose objective is after-cost portfolio performance. Using our framework, future work can add further predictors for volume that may provide even larger economic benefits than we show here. However, our framework provides a way to assess those predictive contributions in economic terms.

[14]Triple witching happens on the third Friday in March, June, September, and December. Double witching is on the third Friday in the other eight months. Russell index re-balancing is on the fourth Friday of June when the Russell 1000, Russell 2000, Russell 3000, and other Russell indexes are reconstituted, and has "often been one of the highest-volume trading days of the year" for the exchange, due to index-tracking funds adjusting their holdings to reflect the updates. References: https://www.nasdaq.com/articles/the-powerful-impact-of-triple-witching-2021-06-10, and https://www.nasdaq.com/articles/2023-russell-rebalancing:-what-you-need-to-know

3 Volume Prediction from a Statistical Perspective

We start with a statistical prediction of daily trading volume using various subsets of predictors and a variety of methods, including machine learning techniques.

3.1 Prediction methods

We run predictive regressions of $\widetilde{\eta}$ (changes in daily dollar volume) on a set of predictors, X, in the training sample panel to estimate the models.[15] We compare linear models (ols), with neural networks (nn) that allow for non-linear transformations and complex interactions, as well as recurrent neural networks (rnn) that, in addition, allow for state variables to incorporate time series dynamics. The simplest baseline is predicting $\widehat{v}_{i,t} = [\text{ma}_5]_{i,t}$, or in other words, $\widehat{\eta}_{i,t} = 0$. (The "hat" denotes predicted values.)

The neural network implementation is kept simple, standard, and fixed throughout the paper in order to facilitate transparency. The network architecture has three fully-connected hidden layers of 32, 16, and 8 ReLU nodes, respectively, and one linear output node. The size of the input layer is the number of predictors supplied.

Recurrent neural network architecture is particularly appealing for this application as it is designed to capture time-series dynamics. A rnn is analogous to state space models like GARCH, in which the forecast $\widehat{\eta}_{i,t}$ is not only a (non-linear) function of the concurrent predictors $X_{i,t}$, but also of a set of state variables that is the output of the network applied to the previous data point $\{i, t-1\}$. That is, $(\widehat{\eta}_{i,t}, state_{i,t}) = rnn(X_{i,t}, state_{i,t-1})$, where rnn represents the neural network function and *state* are the state variables.[16]

[15] As explained above, predicting $\widetilde{\eta}$ or $\widetilde{v}$ are essentially the same: predicting $\widetilde{\eta}$ as $nn(X)$ is just predicting $\widetilde{v}$ as $nn(X) + \text{ma}_5$, where ma_5 is one of the predictors. From the machine learning perspective, this is implementing a simple residual connection (ResNet; see He et al. 2016).

[16] The *recurrent* neural network processes data sequentially, and recursively passes the state variables to the next time period. Essentially, rnn extracts predictive information from concurrent and lagged predictors $X_{i,t}, X_{i,t-1}, X_{i,t-2}, \ldots$, in contrast to a nn that uses only the concurrent predictors but nothing from the past. Although $X_{i,t}$ contains moving averages of $\widetilde{v}_{i,t-1}, \widetilde{v}_{i,t-2}, \widetilde{v}_{i,t-3} \ldots$, for example, the way such lagged information enters the model without an rnn architecture is highly restrictive. With rnn, the model can "learn" flexible dynamics, where time-series dependencies are parameterized by trainable network weights. We implement the rnn with the popular and standard lstm (long short-term memory) architecture. The number of layers and neurons are kept the same as nn, but the total number of parameters increases by four times (due to the flow of lagged information). Specifically, the bottom hidden layer in the aforementioned 3-layer network is upgraded to an lstm layer with 32 hidden states and cell states, with the rest of the two layers unchanged. Lstm is a standard and popular type of

Table 1: Prediction accuracy

cum predictor sets	tech	+fund-1	+fund-2	+calendar	+earnings
# predictors	8	14	161	165	175
Panel A: R^2 relative to $\widetilde{\eta} = \widetilde{v} - \text{ma}_5$					
ma_5	0				
ols	12.09	12.26	12.27	14.85	15.99
nn	14.31	14.90	14.42	17.13	18.45
rnn	15.80	16.25	15.47	18.12	19.86
Panel B: number of parameters					
ma_5	0				
ols	9	15	162	166	176
nn	961	1,153	5,857	5,985	6,305
rnn	6,049	6,817	25,633	26,145	27,425

Each row represents a prediction model, and each column cumulatively adds to the set of predictors. The R^2 is calculated with ma_5 as the benchmark: $R^2 = 1 - \text{MSE}/\text{avg}(\widetilde{v} - \text{ma}_5)^2$, where $\text{MSE} := \text{avg}(\widetilde{v} - \widehat{v})^2 = \text{avg}(\widetilde{\eta} - \widehat{\eta})^2$ and $\text{avg} := \frac{1}{|\text{OOS}|}\sum_{i,t\in\text{OOS}}$ is the OOS average. Each reported R^2 value is the average across five independent runs initialized with different random seeds to ensure the results' robustness and reproducibility. Panel B reports the number of parameters.

Appendix A.1 contains other details on implementing the machine learning methods, including the optimizer, training scheme, and infrastructure. We do not tune or optimize the hyperparameters, the architecture, or the training scheme to improve the results.

3.2 Prediction results

Table 1 reports the OOS prediction accuracy of each method. We cumulatively add sets of predictors in the columns from left to right to highlight the prediction improvement from using larger data sets. The rows correspond to different prediction methods. Panel A reports OOS prediction accuracy in terms of the R^2, where the benchmark is the five-day moving average of volume.[17] Panel B reports the number of parameters estimated in each configuration.

Volume changes are highly predictable. The most sophisticated model using all predictors can

rnn with four specific internal mechanisms, or gates, that control the flow of information from both the short- and long-term past (Hochreiter and Schmidhuber, 1997). See Kelly and Xiu (2023) for a general reference and Appendix A.1 for our specifications.

[17] In addition to controlling for the trailing mean in the prediction target, Appendix B.2 considers an alternative predictive specification that standardizes $\eta_{i,t}$ by a rolling standard deviation. The rationale is that the standardized volume shock may be more conducive to forecasting. We find the predictability remains largely similar. Appendix Table B.2 recasts the R^2's in terms of the explained percentage of the total variation of $\widetilde{v}$.

predict nearly 20% of future variation in daily trading volume innovations. In comparison, daily stock returns are barely predictable with tiny OOS R^2, even with state of the art models and predictors, and monthly returns are only slightly predictable with low single digit R^2's (Gu, Kelly, and Xiu, 2020).

Adding more predictors improves accuracy. Using all 175 predictors increases the R^2 by more than three percentage points compared to just the eight technical signals. The exception is that adding the large set of fundamental signals (fund-2) makes the methods (even ML methods) perform a little worse. This may be due to overfitting when the number of features increases and because we do not use regularization techniques to try to mitigate overfitting. The predictors associated with expected returns do not necessarily work for predicting volume. Market-wide calendar events are quite effective in capturing volume changes, however. Scheduled earnings announcements also add a sizable gain in prediction accuracy.

3.2.1 Feature attribution

We also assess the importance of each variable for prediction using Shapley values borrowed from cooperative game theory. Specifically, we define SHAP (SHapley Additive exPlanation) as follows. Given a trained model $f : \mathcal{X} \to \mathbb{R}$, an instance $x \in \mathcal{X}$, and a background (baseline) distribution for X, SHAP seeks attributions $\{\phi_i(x)\}_{i=1}^{M}$ over feature indices $\mathcal{N} = \{1, \ldots, M\}$ such that

$$f(x) - \mathbb{E}[f(X)] = \sum_{i=1}^{M} \phi_i(x) \quad \text{(local accuracy / additivity)}. \tag{1}$$

The baseline $\mathbb{E}[f(X)]$ anchors explanations, where changing the background changes absolute magnitudes but preserves the additive identity.

For feature $i \in \mathcal{N}$, the Shapley value averages i's marginal contribution over all coalitions $S \subseteq \mathcal{N} \setminus \{i\}$:

$$\phi_i(x) = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!\,(M - |S| - 1)!}{M!} \Big(v(S \cup \{i\}) - v(S) \Big), \tag{2}$$

where the value function $v(S)$ is the model output with only S "present." In ML settings a standard

choice is the conditional expectation

$$v(S) = \mathbb{E}[f(X) \mid X_S = x_S]\,. \tag{3}$$

Using marginal expectations $v(S) = \mathbb{E}[f(X_S = x_S,\, X_{\bar{S}} \sim \text{bg})]$ is also common. The choice affects how credit is split under feature correlation.

In addition, we can group the variables and calculate SHAP values at the group level. Suppose the feature set is partitioned into G disjoint groups (subgroups), $\mathcal{N} = \bigsqcup_{g=1}^{G} \mathcal{G}_g, \quad \mathcal{G}_g \subseteq \mathcal{N}$, then the SHAP (signed) attribution for group g is simply the sum of member feature SHAP values:

$$\Phi_g(x) \;=\; \sum_{i \in \mathcal{G}_g} \phi_i(x). \tag{4}$$

By additivity,

$$f(x) - \mathbb{E}[f(X)] \;=\; \sum_{g=1}^{G} \Phi_g(x). \tag{5}$$

Group attributions preserve the SHAP additive explanation property and are exact (up to numerical approximation) if the underlying ϕ_i are exact.

Figure 2 Panel A plots the SHAP values by feature category. The tech signals have the greatest importance followed by calendar effects and earnings dates. Panel B reports the SHAP values for the top 20 variables based on feature importance, excluding technical predictors (and for the rnn model). The variables with the most important predictive content are triple witching dates, market capitalization, and earnings dates.

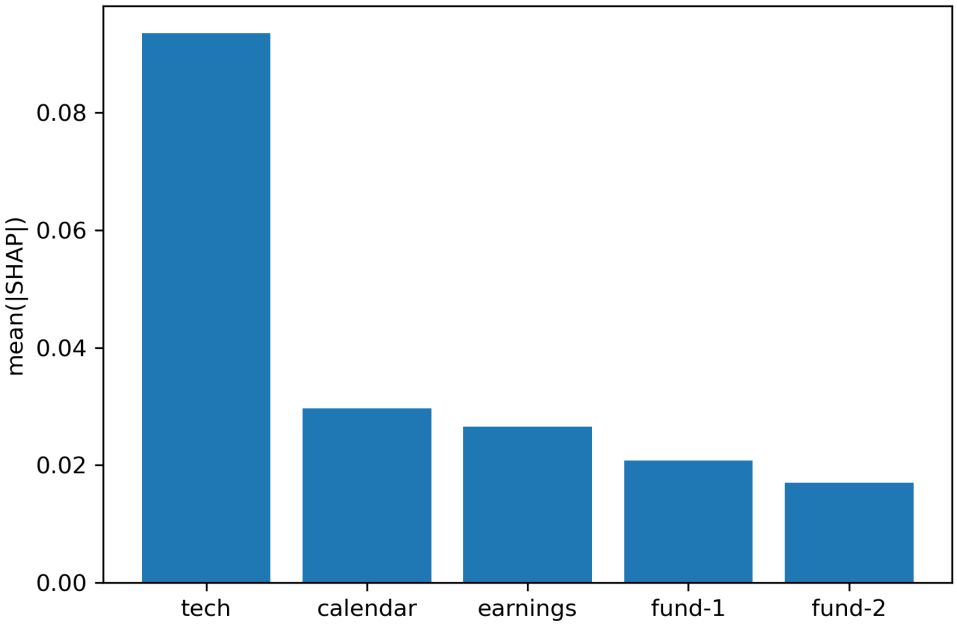
3.2.2 Summary

The results show that machine learning is also useful for prediction, and that complexity has its virtue in the context of predicting volume. The prediction accuracy of the rnn is better than the nn, which in turn is better than ols, uniformly across each configuration of included predictors. Even simple ols models can deliver double-digit R^2, especially when using the largest set of predictors.[18]

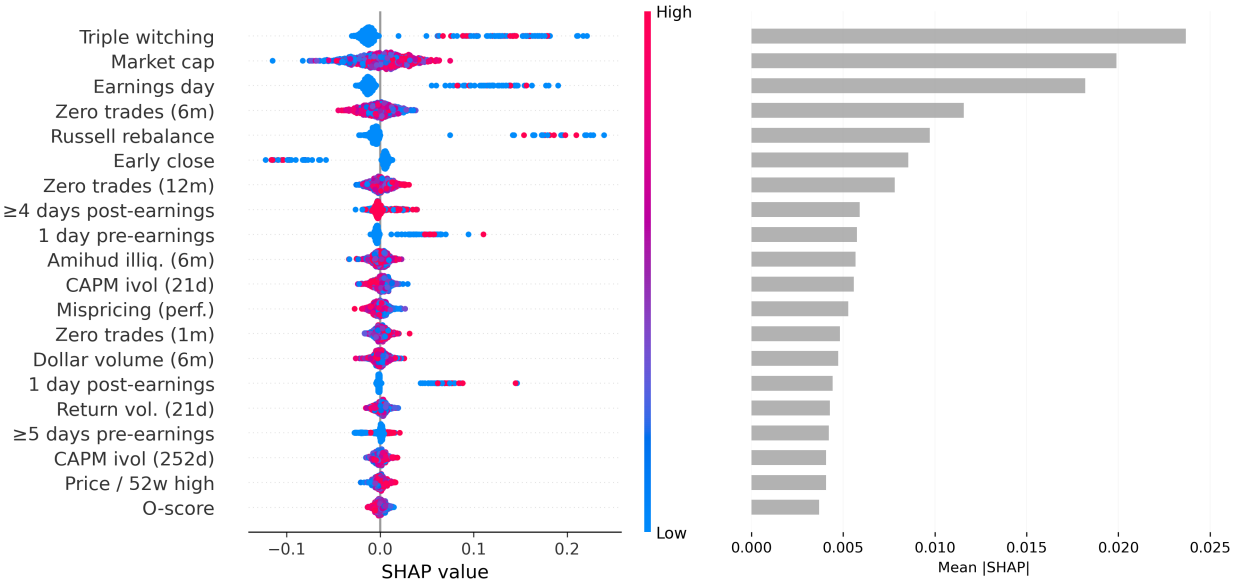
[18] We experimented with regularization on the linear model (lasso and ridge regressions), and did not find significant improvements.

Figure 2

Panel A: Feature group importance



Panel B: Individual feature importance, top 20 excluding technical predictors



Panel A reports the feature group importance measured by mean absolute SHAP values. This is computed on the test set for the rnn model. Panel B reports the feature importance for the top 20 individual features measured by mean absolute SHAP values, excluding technical predictors. Scatter points represent individual instances, colored by feature value (blue: low, red: high), horizontal axis is SHAP value (contribution to volume forecast). Bar plot is mean absolute SHAP value across instances. Reported values are for the rnn model.

Panel B shows the improved prediction accuracy is achieved through a significant increase in the number of parameters, a measure of model complexity. Appendix A.1 shows the computational costs of the complex models are higher but manageable.

Appendix B.3 considers volume predictability in different firm size groups. Larger firms have higher prediction accuracy than smaller firms, but the overall patterns of predictability across methods are robust in each size sub-sample. The R^2's evaluated among the mega firms are roughly twice those of the nano firms. Smaller firms have a greater magnitude of unexpected trading volume shocks that are harder to predict. Small firms are volatile and have low trading volume, hence unexpected events that give rise to volume spikes may have bigger impact for these firms. This finding indicates that in addition to small firms being less liquid on average, their liquidity is also less predictable and more volatile. Thus, their trading costs are less predictable.[19] This characteristic of small firms will be taken into account when we convert volume forecasts into their economic marginal contribution for portfolio optimization. In the next section, we form an economic objective based on the portfolio problem to assess the value of volume prediction.

4 The Economic Value of Volume Forecasting

We assess the economic value of volume forecasting by converting volume forecasts into an economic outcome.

4.1 Volume weighted average price (VWAP)

We start with a simple exercise using VWAP, which represents the best available traded price and is a benchmark for best execution or market making activity. Converting our volume forecasts into VWAP forecasts gives it some economic content, but in a limited way. Here, we only use the volume forecasts and take prices as given.

Rather than look at intra-day volume, since we are forecasting daily volume, we define the daily

[19] We also examine whether firms of different size groups should be modeled differently by implementing a mixture of experts method. While this method is somewhat beneficial for the linear model, it is not for the neural networks.

dollar-weighted average trade price as

$$P_{i,t}^{\text{avg}} \;=\; \frac{D_{i,t}}{Q_{i,t}}, \qquad \text{for } D_{i,t} > 0,\; Q_{i,t} > 0, \tag{6}$$

where $D_{i,t}$ is realized daily dollar volume and $Q_{i,t}$ is realized daily share volume. We then examine a trading horizon H (e.g., weekly or monthly) to assess VWAP prediction coming from our daily volume forecasts.

The realized VWAP for stock i over horizon H is

$$\text{VWAP}_{i,H} \;=\; \frac{\sum_{t\in H} D_{i,t}}{\sum_{t\in H} Q_{i,t}} \;=\; \frac{\sum_{t\in H} D_{i,t}}{\sum_{t\in H} D_{i,t}/P_{i,t}^{\text{avg}}}. \tag{7}$$

Given a forecast of dollar volume $\widehat{D}_{i,t}$, we form the forecast horizon VWAP by translating forecast dollars into implied shares using the realized $P_{i,t}^{\text{avg}}$:

$$\widehat{Q}_{i,t} \;=\; \frac{\widehat{D}_{i,t}}{P_{i,t}^{\text{avg}}}, \qquad \widehat{\text{VWAP}}_{i,H} \;=\; \frac{\sum_{t\in H} \widehat{D}_{i,t}}{\sum_{t\in H} \widehat{Q}_{i,t}} \;=\; \frac{\sum_{t\in H} \widehat{D}_{i,t}}{\sum_{t\in H} \widehat{D}_{i,t}/P_{i,t}^{\text{avg}}}. \tag{8}$$

We then assess the accuracy of our volume forecasts through the accuracy of the H-horizon VWAP as an economic indicator of how much volume prediction matters from an approximate market-making cost. We define the relative absolute error (RAE) of VWAP as,

$$\text{RAE}_{i,H} \;=\; \frac{\left|\widehat{\text{VWAP}}_{i,H} - \text{VWAP}_{i,H}\right|}{\text{VWAP}_{i,H}}. \tag{9}$$

If $\mathcal{H}_i$ is the set of horizons for stock i (e.g., all months), the average error for stock i is

$$\overline{\text{RAE}}_i \;=\; \frac{1}{|\mathcal{H}_i|} \sum_{H\in\mathcal{H}_i} \text{RAE}_{i,H}, \tag{10}$$

which is averaged across stocks, both equal- and volume-weighted, and reported in Table 2.

The first two columns of Table 2 report the equal- and volume-weighted average relative absolute error across all stocks at a monthly horizon and the last two columns at a weekly horizon, for

Table 2: VWAP forecasting accuracy (OOS RAE in percent)

	Monthly VWAP		Weekly VWAP	
model	EW RAE	VW RAE	EW RAE	VW RAE
ma_5	1.33	0.58	0.77	0.37
ols	1.13	0.48	0.65	0.31
nn	1.04	0.47	0.63	0.30
rnn	1.03	0.43	0.63	0.30

Reported are the equal-weighted (EW) and volume-weighted (VW) relative absolute error in forecasting VWAP across stocks. The unit of RAE numbers is in percent. The ols, nn, and rnn models use the full set of predictors.

various models. Consistent with our previous results, the rnn and nn models produce the best OOS forecasts, reducing RAE from 1.33% using just a five-day moving average of volume ma_5 to 1.03% using a rnn for EW averages, and from 0.58 to 0.43 for VW averages. The impact is even larger when we look across different sized stocks, where the errors can be 4-5% or more for micro caps, for instance (results reported in Appendix Table D.6).

A market maker or best execution trader could benefit by as much as 0.30% to almost 5% in execution costs from more accurate volume forecasts relative to a simple five-day moving average of volume according to our VWAP analysis. However, while this analysis provides a glimpse of economic content to volume forecasts, it is limited in scope. VWAP simply represents the best price available to trade at the quantities traded, but does not account for any further market impact, and therefore does not vary with the size of the trade or the size of the portfolio. It does not allow for an endogenous portfolio choice response. In the next subsections, we develop a portfolio choice framework to provide a richer analysis of the value of volume prediction.

4.2 Trading volume alpha

To better assess the value of volume prediction to asset pricing, we set up a portfolio choice problem that features a tradeoff between tracking a target portfolio versus minimizing trading costs. While this framework provides a richer analysis that endogenizes trading and varies by portfolio size, it does require some structural assumptions, which we lay out here.

Our portfolio optimization targets the net-of-cost performance of the portfolio, expressed as

$$\widetilde{\text{P\&L}}_t = \sum_i x_{i,t}\widetilde{r}_{i,t} - \sum_i TradingCost(x_{i,t}, x^0_{i,t}, \widetilde{\Omega}_{i,t}) + Ar_{\text{f},t}, \tag{11}$$

where $\widetilde{r}_{i,t}$ is the return of stock i on day t in excess of the risk-free rate $r_{\text{f},t}$, A is the portfolio size (AUM), and $\widetilde{\Omega}_{i,t}$ is a vector of attributes, whose values are unknown at the time of trading, that are related to the trade itself, the characteristics of the asset traded, and the market conditions under which the asset is traded. Importantly, the portfolio manager chooses the dollar positions $x_{i,t}$ before trading, asset, and market conditions (represented by $\widetilde{\Omega}_{i,t}$) are realized, given the starting positions $x^0_{i,t}$ inherited from the previous day as well as the available information set $\{X_{i,t}\}$, for asset i at time t.[20]

Much of the asset pricing literature focuses on forecasting $\widetilde{r}_{i,t}$, with all of the documented challenges of doing so. Here, we shift the focus to "trading costs" as a potential new avenue for building more efficient portfolios, which has its own set of (new) challenges. To simplify matters and as a starting point to quantify the economic benefits of predicting a non-return variable, we examine trading volume as the key element in $\widetilde{\Omega}_{i,t}$ that affects trading costs. Of course, there are many attributes that affect trading costs, but our goal here is to focus singularly on trading volume to illustrate how we can translate a non-return prediction problem into portfolio economics.

Importantly, we take the before-cost portfolio as given and do not try to optimize the gross-of-cost return on the portfolio, but rather assume it is the optimal portfolio in the absence of costs. This assumption turns the problem into a tracking error problem that allows us to set aside the return prediction problem to focus on the improvement afforded by the volume prediction problem alone. We call this improvement "trading volume alpha."

The incentive to trade is modeled with an objective that penalizes the tracking error toward a target portfolio. The tracking target taken as given implies the key choice is whether to trade aggressively toward the target or passively let it drift away from the target to avoid trading costs. The optimal balancing point depends on the volume forecast, the trading costs associated with it,

[20] When implementing the portfolio, the starting position is inherited from the position on $t-1$ and mechanically adjusted by the raw return: $x^0_{i,t} := x_{i,t-1}\,(1 + r_{\text{f},t-1} + \widetilde{r}_{i,t-1})$.

and the cost of not trading or incurring tracking error. This framework is quite general and applies to any specific tracking target. For example, in addition to thinking of the target portfolio as being one based on return and risk forecasts with the objective of maximizing its Sharpe ratio, the tracking target could be a benchmark index, or a hedge (for a derivative position, investment exposure, consumption exposure, etc.), or an M&A deal, arbitrage trade, accumulating a targeted position for shareholder voting, or meeting an investment constraint (e.g., ESG or tax considerations). Modeling the problem this way allows for many potential applications, where we can evaluate whether more accurate volume forecasts can translate into better trading execution, and quantify the value of those forecasts in various contexts. We will do so for two investment applications in the next section, but there is a wide range of potential targets.

4.3 Tracking error and tcost optimization

With the tracking target given, the tracking error objective is modeled with a simple mean-variance portfolio optimization framework. Suppose the portfolio manager maximizes a mean-variance certainty equivalent of the portfolio P&L, adjusted for trading costs:

$$\sum_i x_{i,t} m_{i,t} - \frac{\gamma}{2A} \sum_{i,j} x_{i,t} x_{j,t} \sigma^2_{ij,t} - \sum_i TradingCost_{i,t} + Ar_{\mathrm{f},t}, \tag{12}$$

where the first and second moments of returns are $m_{i,t}$ and $\sigma^2_{ij,t}$ and are given, and the risk aversion coefficient γ is explicitly adjusted by the AUM, A.[21] Assuming $\mathbb{V}\mathrm{ar}_{t-1}\widetilde{r}_{i,t} = \sigma^2 \quad \forall i$, with zero covariances, the objective function is then

$$-\frac{\gamma\sigma^2}{2A} \sum_i \left(x_{i,t} - \frac{A}{\gamma\sigma^2} m_{i,t} \right)^2 - \sum_i TradingCost_{i,t} + \left(Ar_{\mathrm{f},t} + \frac{A}{2\gamma\sigma^2} \sum_i m^2_{i,t} \right). \tag{13}$$

The task is to balance the tradeoff between the first term, which is the tracking error penalty as the result of the mean-variance optimization, and the second term, the transaction cost, to be detailed below. The third term can be ignored in the optimization since it is irrelevant to the x choices.

[21] The risk aversion coefficient is explicitly adjusted by A such that the before-cost Markowitz optimal dollar position, x^*, scales up with A. Eq. 11 assumes a simple situation where the AUM is constant, assuming immediate P&L payout.

The tracking error penalty is quadratic,

$$TrackingError_{i,t} := \frac{1}{2}\mu(x_{i,t} - x^*_{i,t})^2, \tag{14}$$

where the target x^* is the before-cost optimal portfolio position, which increases in the asset's return expectations as well as the total portfolio size A. Parameter $\mu := \frac{\gamma\sigma^2}{A}$ controls the strength of the tracking error penalty. In later empirical analyses, we do not calibrate μ from risk coefficients but instead treat it as a hyperparameter and tune the optimal μ under various AUM levels according to investment performance. However, the qualitative relationship is preserved – a larger investor penalizes tracking error (measured in dollars) less, meaning they trade less aggressively toward the target in general.[22]

Trading costs are modeled as

$$TradingCost_{i,t} := \frac{1}{2}\widetilde{\lambda}_{i,t}(x_{i,t} - x^0_{i,t})^2, \tag{15}$$

where x^0 is the starting position. We specify $\widetilde{\lambda}$ as a function of volume: $\widetilde{\lambda} = 0.2/\widetilde{V} = 0.2\exp(-\widetilde{v})$, following Frazzini, Israel, and Moskowitz (2018). Underlying the quadratic functional form, it is assumed that the price impact is linear in the trade's size relative to the volume of the day $PriceImpact = 0.1\frac{x-x^0}{\widetilde{V}}$ (Kyle, 1985). For example, buying (or selling) 10% of the daily volume would move the price by 1% (or −1%). The trading cost is the price impact multiplied by the dollar trade size: $TradingCost = PriceImpact \cdot (x - x^0)$.[23] Again, the aim here is to translate volume forecasts into an economic object, in this case tcosts. We are ignoring other variables, like

[22] Our tracking error model assumes that trading costs (and hence volume) are independent of returns/alpha and that tracking error is specified as simply the difference in weights and not the covariance of returns with the before-cost portfolio. The difference in weights is a more direct and simpler approach to measuring tracking error and is more appropriate for applications like voting shares or activist investing. For other applications like benchmarking to a portfolio, the covariance measure is also appropriate, but the difference in portfolio weights also solves this problem. For these reasons we adopt the portfolio weight difference specification. We are also ignoring dynamic effects. However, our empirical trading experiments will take some of these into account, to the extent they show up in the data.

[23] For simplicity, we assume away cross-impact on $\widetilde{\lambda}$ from related stocks as well as other determinants of $\widetilde{\lambda}$. Also, other functional forms for $\widetilde{\lambda}$, such as the square-root form often shown empirically, $\widetilde{\lambda} = \frac{0.2}{\sqrt{\widetilde{V}}} = 0.2\exp(-\frac{1}{2}\widetilde{v})$, (Frazzini, Israel, and Moskowitz, 2018) can also be used. In this case, all the analyses carry through but $\widetilde{v}$ will be twice as large. We stick with the linear specification consistent with theory (Kyle, 1985) and empirical evidence on the unexpected component of trading volume (see Frazzini, Israel, and Moskowitz 2018).

volatility, that also impact tcosts and we are not varying λ by stock or by time. Presumably, these additions would provide more accurate forecasts and more precise measures of true costs that would improve portfolio execution. However, we wish to simplify the analysis for better intuition and presentation, and we will show that even this simple translation of volume to costs has significant economic content and can improve decision making. A more complicated and accurate framework may have even stronger economic impact.

In summary, the aggregate tracking error optimization problem is,

$$\min_{\{x_{i,t}\}} \sum_{i,t} \left(TrackingError_{i,t} + TradingCost_{i,t} \right). \tag{16}$$

Central to our paper, volume $\widetilde{v}_{i,t}$, and therefore price impact, are not known when choosing $x_{i,t}$ and must be forecasted. In addition, $x^0_{i,t}$ and $x^*_{i,t}$ are taken as given, which makes the problem $\{i,t\}$-separable. In reality, $x^0_{i,t}$ is affected by the choice on the previous day. While we do not model such dynamics in the trading experiments (Section 6), we consider dynamics by evaluating the traded portfolios recursively.[24]

With all of these assumptions, the problem becomes[25]

$$\min_{x \in \sigma(\mathcal{X})} \mathbb{E}\left[\frac{1}{2}\widetilde{\lambda}(x - x^0)^2 + \frac{1}{2}\mu(x - x^*)^2 \right].$$

To implement this problem empirically, we first normalize the problem by the target trade size $x^* - x^0$ so that we analyze the loss for a one-dollar target trade. The problem scales quadratically. For example, an $x^* - x^0 = \$1{,}000$-dollar trade task will incur 10^6 times the loss of a \$1-trade. We

[24] One other caveat/assumption is that we are not considering trading permanently impacting the price of a security, which would create a feedback loop that alters the optimal target itself, eroding alpha of those positions. First, for a small investor, this dynamic can be effectively ignored since they are unlikely to move the price. Second, even for large investors, most of the price impact is temporary and not permanent (Frazzini, Israel, and Moskowitz, 2018), but to the extent there is some permanent impact, this framework may be less suitable for really large investors.

[25] Here, "$x \in \sigma(\mathcal{X})$" restricts x as a random variable that depends only on predictive information available at the time of the choice, with $\mathcal{X}_{i,t} := [X_{i,t}, X_{i,t-1}, X_{i,t-2}, \dots]$. This unconditional expectation minimization is equivalent to solving $\min_{x \in \mathbb{R}} \mathbb{E}\left[\frac{1}{2}\widetilde{\lambda}(x - x^0)^2 + \frac{1}{2}\mu(x - x^*)^2 \middle| \mathcal{X} \right]$ for each $\mathcal{X}$ realization.

define the choice variable trading rate $z := \frac{x-x^0}{x^*-x^0}$, where the minimization objective becomes

$$\frac{1}{2}\widetilde{\lambda}(x-x^0)^2 + \frac{1}{2}\mu(x^*-x)^2 = \frac{1}{2}\left(x^*-x^0\right)^2\left(\widetilde{\lambda}z^2 + \mu(1-z)^2\right). \tag{17}$$

Since the factor, $\left(x^*-x^0\right)^2$, does not matter for the choice of z, the economic loss is defined as

$$loss^{\text{econ}}(\widetilde{v}, z; \mu) := \widetilde{\lambda}z^2 + \mu(1-z)^2, \tag{18}$$

and the normalized problem becomes

$$\min_{z\in\sigma(\mathcal{X})} \mathbb{E}\left[loss^{\text{econ}}(\widetilde{v}, z; \mu)\right]. \tag{19}$$

Separating x^*, x^0 affords many conveniences. It means the core problem is independent of the target strategy or fund size. It allows us to look at each i,t observation independently in a volume prediction setting. After the prediction task is done, we evaluate the investment performance under various pre-specified target strategies (x^*) with different AUM levels.

4.4 The optimal policy without forecast error

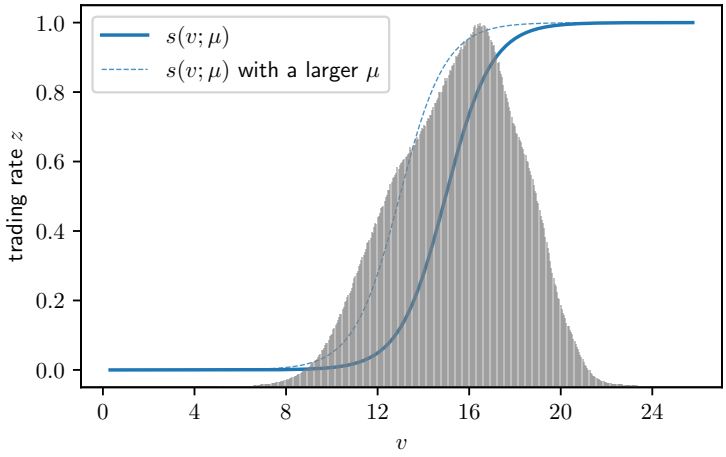
Suppose we have perfect foresight on volume, $\widetilde{v} = v$, (no forecast error) then the optimal policy is,

$$s(v;\mu) := \arg\min_z loss^{\text{econ}}(v, z; \mu) = \frac{\mu}{\mu+\lambda} = \frac{1}{1+\exp\left(-v+\log 0.2-\log\mu\right)}. \tag{20}$$

We plot this function in Figure 3, with the trade rate z on the vertical axis and trading volume v on the horizontal axis. It is a sigmoid function with a horizontal offset determined by μ. The optimal trading rate z ranges from 0 to 1 as v increases, but converges to 1 (0) at large (small) values of v. The optimal dollar position choice is $x = x^0 + s(v;\mu)(x^*-x^0)$. This is the finding of Gârleanu and Pedersen (2013), where the optimal strategy should "trade partially toward the aim."[26] In their setting, the trading rate is a fixed constant. Here, it is still irrelevant to either the

[26]The other finding in Gârleanu and Pedersen (2013) is that the optimal strategy should also "aim in front of the target." This dynamic effect is abstracted away in our problem since we are considering the $\{i,t\}$-separable

Figure 3: The policy function (s) that maps log volume (v) to trading rate (z)



The solid curve uses a μ value relevant for \$1b AUM. The dashed curve changes to a greater μ (AUM = \$100m), increasing the trade rate z across the spectrum. The background is the histogram of actual volume $\widetilde{v}$ from Figure 1, to show that over most of the distribution of $\widetilde{v}$, $\widetilde{z}$ will be between 0 and 1 given the μ's associated with \$1b and \$100m funds.

trading target or the starting point (x^*, x^0), but importantly, it depends on the volume prediction, v. Instead of assuming liquidity as a constant known by the agent, the innovation and emphasis of our paper is that *volume prediction* (and prediction error) alters the tradeoff between the cost and benefit of trading, where more accurate predictions lead to more efficient portfolio implementation, with z varying with forecasted volume.

Additionally, the optimal z does not explicitly depend on the scale of the fund. If the AUM doubles, and both x^* and x^0 double, the optimal z remains the same, while the dollar position choice x doubles. However, a smaller fund will find a larger μ more relevant for their investment performance optimization. In that case, they will trade more aggressively uniformly across v, as illustrated by the upward shift of the dashed curve in Figure 3, although the magnitude of the shift is not constant. For example, at very low (high) trading volume, both large and small managers will trade very little (aggressively) implying z close to zero (one). However, over the majority of the distribution of actual volume (plotted on the graph), small managers with larger μ will trade more aggressively (higher z), with differences in z being significant over most of the volume distribution.

optimization. One interpretation is that x^* is already the aim that is in front of the target that implicitly embeds the dynamic effect. Our empirical implementation will implicitly take this into account.

5 Machine Learning for the Economic Value of Volume Prediction

We develop empirical methods to construct the policy of choosing z given $\mathcal{X}$.

5.1 Statistical versus economic learning for volume prediction

We consider two ways of approaching the portfolio optimization problem. The first finds a statistically efficient prediction of volume and then plugs that volume forecast into the optimal trading policy $s(v;\mu)$. This is the classic approach to implementing economic models, which we call "statistical" learning. The second approach takes a more direct route that incorporates the economic objective into the prediction problem, where we predict the trading rate z (as a function of conditioning information) directly to minimize the economic loss and then back out the implied volume for that trade rate z. Choosing z implies v because there is a direct mapping from z to v, but the mapping comes from the optimization objective rather than a statistical criterion. We term this approach "economic" learning. The two approaches can yield different results because unlike a statistical loss function, the economic loss function penalizes volume forecast errors asymmetrically, where overestimating volume is more costly than underestimating volume under some regions of the volume distribution. In such a case, the model optimized for the economic goal will be more conservative at the expense of compromising minimizing squared errors.

- Approach 1, statistical learning:

  Step 1: run machine learning regressions of $\widetilde{v}$ onto $\mathcal{X}$ in the training sample as in Section 3

  $$v^*(\cdot) = \arg\min_{v(\cdot)} \sum_{i,t\in\text{train}} (\widetilde{v}_{i,t} - v(\mathcal{X}_{i,t}))^2. \tag{21}$$

  Step 2: plug the OOS predictions $\widehat{v}_{i,t} := v^*(\mathcal{X}_{i,t})$ into policy equation (20) to trade $\widehat{z}_{i,t} = s(\widehat{v}_{i,t};\mu)$.

- Approach 2, economic learning:

parameterize z as a neural network, optimize the economic objective in the training sample

$$z^*(\cdot) = \arg\min_{z(\cdot)} \sum_{i,t\in\text{train}} loss^{\text{econ}}(\widetilde{v}_{i,t}, z(\mathcal{X}_{i,t}); \mu) \tag{22}$$

and trade $\widehat{z}_{i,t} = z^*(\mathcal{X}_{i,t})$ in the testing sample.

The difference between the two approaches is the different loss functions deployed in penalizing volume prediction errors. A trading action $\widehat{z} = z(\mathcal{X})$ implies an underlying volume forecast $\widehat{v} = s^{-1}(\widehat{z}; \mu)$, and equivalently the economic loss can be written as a function of the z-implied v instead of z itself: $loss^{\text{econ}}_{\text{vv}}(\widetilde{v}, v; \mu) := loss^{\text{econ}}(\widetilde{v}, s(v; \mu); \mu)$. Hence, the economic learning approach is equivalent to first solving

$$\min_{v(\cdot)} \sum_{i,t\in\text{train}} loss^{\text{econ}}_{\text{vv}}(\widetilde{v}_{i,t}, v(\mathcal{X}_{i,t}); \mu) \tag{23}$$

followed by the $s(\,\cdot\,; \mu)$ transformation (which is also required in the first approach).

Given that the two approaches are only different in the loss functions, we compare $loss^{\text{econ}}_{\text{vv}}$ with the least squares loss function, $loss^{\text{ls}}(\widetilde{v}, v) := \frac{1}{2}(v - \widetilde{v})^2$, used in statistical predictions. To understand how the two approaches differ, first note that the two functions are the same for the smallest possible loss being attained, which is when the forecast v exactly equals the target, $\widetilde{v}$. In empirical experiments, we label the strategy made with perfect foresight "oracle" as the unattainable ideal ($\widehat{v}_{i,t} = \widetilde{v}_{i,t}$), which yields the smallest mean squared error (MSE) and smallest mean economic loss (MEL).

Second, both loss functions monotonically increase as the forecast v deviates away from the true $\widetilde{v}$. Therefore, it is intuitive to think that making forecasts that are close to $\widetilde{v}$ in the least squares sense, will translate to better portfolio implementation as evaluated by the economic loss.[27] However, forecast errors will not guarantee this outcome because of the differences between the two loss functions, in particular the economic loss function being asymmetric in forecast errors at different parts of the volume distribution. From a theoretical perspective, it is well known that the

[27]The above mentioned properties expressed mathematically are $\arg\min_v loss(v, \widetilde{v}) = \widetilde{v}, \forall \widetilde{v}$; and $loss(v, \widetilde{v})$ is increasing in $|v - \widetilde{v}|, \forall v, \widetilde{v}$. Both $loss^{\text{econ}}_{\text{vv}}$ and $loss^{\text{ls}}$ satisfy these properties.

conditional expectation, $\mathbb{E}[\widetilde{v}|\mathcal{X}]$, is the minimizer of $\min_{v\in\sigma(\mathcal{X})}\mathbb{E}\left[loss^{\text{ls}}(\widetilde{v},v)\right]$, so that the statistical learning method recovers the conditional expectation with the neural network tools. However, given a different economic loss function,

Proposition 1. *The least squares minimizer,* $\mathbb{E}[\widetilde{v}|\mathcal{X}] = \arg\min_{v\in\sigma(\mathcal{X})}\mathbb{E}\left[loss^{\text{ls}}(\widetilde{v},v)\right]$*, does not optimize the economic loss minimization problem* $\min_{v\in\sigma(\mathcal{X})}\mathbb{E}\left[loss^{\text{econ}}_{\text{vv}}(\widetilde{v},v;\mu)\right]$.

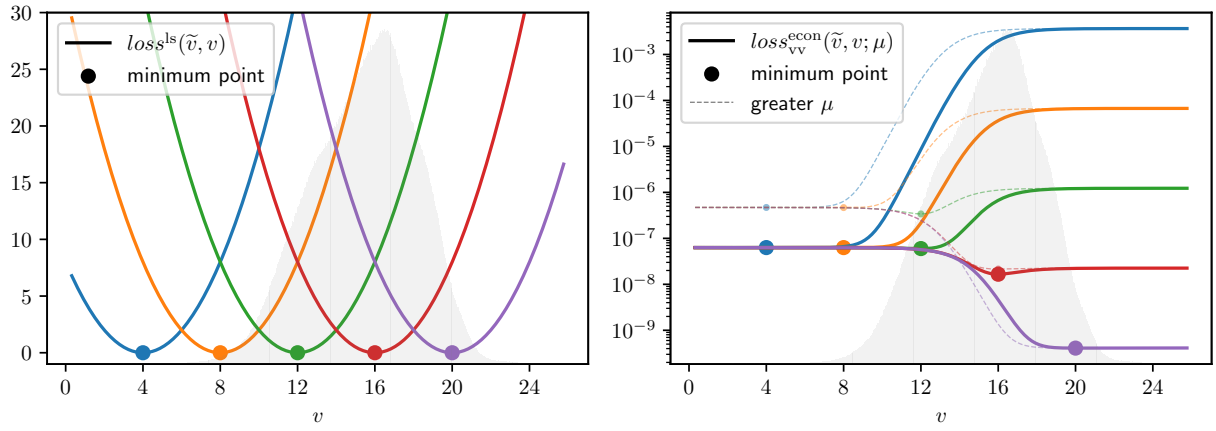
Even with unlimited data, the "perfect" statistical learning would not optimize the economic problem. The theoretical foundation of this claim is that $\mathbb{E}[\widetilde{v}|\mathcal{X}] = \arg\min_{v\in\sigma(\mathcal{X})}\mathbb{E}\left[\phi(\widetilde{v},v)\right]$ *if and only if* the generic loss function ϕ is in the Bregman class (Banerjee, Guo, and Wang, 2005; Patton, 2020). While the least squares loss belongs to the Bregman class, the economic loss function does not (proofs in Appendix C.3).

Figure 4 illustrates the difference between the two loss functions. We pick five different true values for $\widetilde{v} = 4, 8, 12, 16, 20$, and plot the loss curves for a range of volume forecasts $v \in [0, 24]$. The left graph plots the statistical loss relative to the true value for volume. The dots mark the minimum loss where the forecast $\hat{v} = \tilde{v}$ ("the oracle"). Given the quadratic statistical loss function, forecast errors on either side of $\tilde{v}$ generate symmetric losses. The right graph in Figure 4 plots the economic loss relative to volume forecasts.[28] The distinguishing feature of the economic loss is its asymmetric penalty for over or underestimating volume over various regions of the volume distribution. For low actual volume, overestimating volume is extremely costly as it causes much larger price impact. For very high volume, the opportunity cost of not trading aggressively enough is larger than the cost of trading and hence underestimating volume when volume is high is more costly. Comparing the values of v over the actual distribution of realized trading volume (whose histogram is plotted in the background), it appears that overestimating volume is on average more costly than underestimating it over most of the distribution of actual trading volume.

For example, the blue loss curve (actual $\widetilde{v} = 4$) is very high if the forecast is mistakenly large. The economic intuition for why this particular forecasting error is so costly is that it implies trading aggressively (z close to 1) when actual liquidity is low. Analytically, $\lim_{z\uparrow 1} loss^{\text{econ}}(\widetilde{v},z;\mu) = \widetilde{\lambda}$,

[28] In the right panel, the dips around the minimums (indicated by the dots) are too shallow to be noticeable, though analytically they are indeed the minimums. See Appendix C.4 for more details on the local curvature around the minimum points.

Figure 4: Statistical versus economic loss functions



The first panel plots the statistical (least squares) loss function, $loss^{ls}(\widetilde{v}, v)$ and the second panel plots the economic loss function, $loss^{econ}_{vv}(\widetilde{v}, v; \mu)$ from our portfolio optimization framework. We pick five different true values for volume, $\widetilde{v} = 4, 8, 12, 16, 20$, and respectively plot the loss curves for $v \in [0, 24]$. The dots mark the minimums of the loss curves, attained at $v = \widetilde{v}$. The right panel is in the log scale. The solid and dash curves use the μ values in columns 2 and 3 of Table 3 (corresponding to \$1b and \$100m AUM), respectively.

which increases exponentially if $\widetilde{v}$ is small. In contrast, errors in the other direction are much more forgiving (e.g., the purple curve with $\widetilde{v} = 20$).[29] Trading too little when actual liquidity turns out to be ample delivers a loss equal to the opportunity cost of not tracking the target portfolio, which is fixed at μ ($\lim_{z \downarrow 0} loss^{econ}(\widetilde{v}, z; \mu) = \mu, \forall \widetilde{v}$). Thus, for low levels of true volume, overestimating volume is very costly because price impact is unbounded, but for high levels of volume, the economic cost of volume forecast error is relatively small because the opportunity cost of not trading is bounded.

We state this asymmetric property formally in Proposition 4 in Appendix C.4, and provide further analysis. The analytical results rely on the quadratic functional forms assumed in equations (14) and (15). However, the qualitative points carry over to more general settings.

These findings have important implications. Off-the-shelf machine learning tools may not be the most suitable for specific portfolio problems because they minimize statistical error rather than economic error. An altered financial machine learning design that accounts for economic loss can be more effective for solving the portfolio problem. The ranking of forecasts evaluated by the two

[29] Notice the vertical axis is on a log scale, meaning the purple is much flatter compared to blue in terms of the difference of its right and left ends. The same plot in linear scale is in Appendix Figure C.3.

loss functions can be different – a set of forecasts with a smaller sample MSE might have a greater economic loss – something we will see empirically in the next section. Enabling the model to "learn" to be conservative in avoiding overestimating volume, compromising statistical accuracy in favor of minimizing economic losses, may lead to better portfolio outcomes.

We implement the statistical and economic learning approaches with neural networks, given the many benefits of deep learning such as the ability to handle high-dimensional data and non-linear relations. The network architectures are kept the same for both approaches, with the only difference being the loss function used in training.

Recall that the optimization chooses the variable z, the trade rate, which then implies a forecast for v, following equation (18) and illustrated in Figure 3. This takes into account the functional form relating v to z and the economic loss. This is different than just forecasting $\frac{1}{V}$ with a statistical loss function because while $\frac{1}{V}$ is proportional to Kyle's λ, the optimal trade rate, which is what we want from a portfolio optimization perspective is nonlinear in λ ($z = \frac{\mu}{\mu+\lambda}$). Moreover, these functional forms come from theory and embed the economic loss directly into the objective.

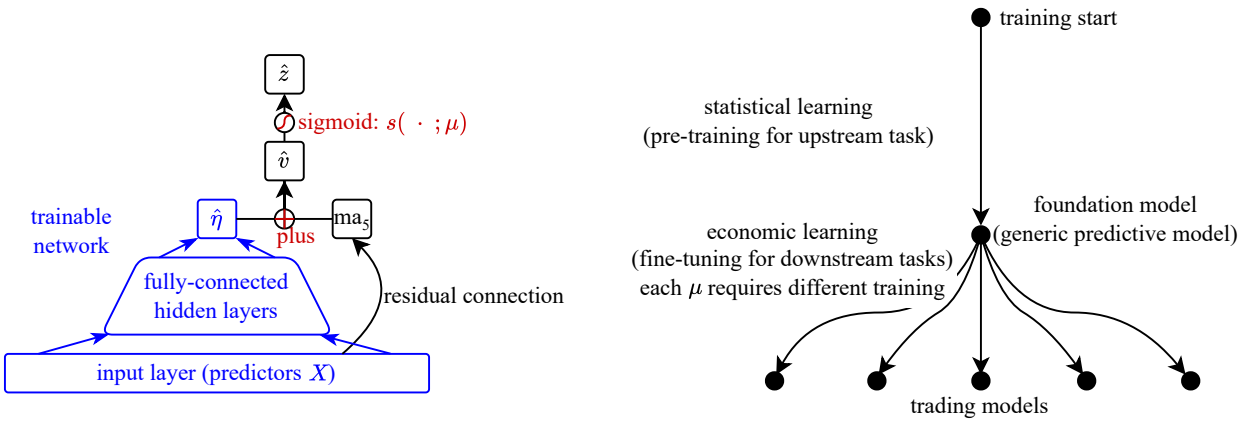
5.2 Transfer learning via pre-training and fine-tuning

We implement a transfer learning paradigm in which the statistical and economic learning models are trained sequentially, as illustrated in Figure 5.

The statistical volume prediction is an upstream task. It learns valuable but generic information on the predictive relationship and serves as the foundation model, using off-the-shelf machine learning programs optimized for this typical task. The background knowledge can be transferred to more specific downstream tasks such as portfolio optimization in our case. The finance-motivated tasks benefit from a good "foundation." Fine-tuning the pre-trained foundation model per the economic loss objective further improves the economic performance. The different specific downstream tasks require separate economic training routines. (The tasks are different because the economic loss function is modulated by μ.) The same pre-trained foundation model serves as the common starting point for different downstream fine-tuning.[30]

[30] An even lower-level downstream task is to make decisions given target position x^*. We do not directly train for

Figure 5: Network architecture and transfer learning procedure



The left illustration shows the network architecture. The blue part "trainable network" is a standard feed-forward neural network, specified with three fully connected hidden layers with 32, 16, and 8 neurons, respectively. The black parts are non-trainable transformations from $\widehat{\eta}$ to log volume prediction $\widehat{v}$ and ultimately trading rate $\widehat{z}$. The recurrent connections in rnn are omitted in this illustration. The right figure illustrates the training procedure for transfer learning. Each dot represents a trained model, i.e., a parameterization of the network. The arrows show economic learning as specific fine-tuning steps based on the common foundation model pre-trained statistically.

Pre-training optimizes a nn or rnn from X to $\widehat{\eta}$ (the blue part in Figure 5 left panel) as described in the previous section. Using the pre-trained network as is, flowing the output $\widehat{\eta}$ through additional transformations, "+ma_5" and "sigmoid $s(\cdot\,;\mu)$" shown in black, the resulting $\widehat{z}$ implements the plug-in step. Taking this foundation model as the initialization, fine-tuning conducts further stochastic gradient descent in the economic loss function evaluated at the same training sample. The resulting fine-tuned network implements the economic learning approach. Only the trainable part (in blue) is updated in fine-tuning. The economic approach only innovates on the loss function, not the network architecture or data. Neural networks tackle the non-linearity not only in the predictive relationship but also in the (marginal) economic loss function. Experiments show fine-tuning requires only a small number of epochs of training to significantly improve OOS economic performance compared to the pre-trained foundation model.[31]

Many other finance problems share a similar structure, in which statistical results are transferred into actionable strategies that are applied towards an economically motivated problem. Markowitz

that, but do evaluate the performance in such trading experiments below.

[31] Alternatively, side-stepping pre-training but directly training for the economic loss from scratch is generally less robust and takes more time (epochs over the sample) to train, probably because the machine learning program is not optimized for such a loss. Not to mention that each μ would require a separate training routine as they do not share the common pre-trained baseline.

portfolio optimization is a classic example. Another example is financial risk management, which relies on volatility forecasts. The transfer learning procedure adopted here provides a unified framework for applying machine learning techniques in these scenarios.[32] The procedure is also similar to how GPT models are "P"re-trained on language representations and applications like ChatGPT are fine-tuned for generating conversational responses or other tasks.

5.3 Economic and statistical prediction results

Table 3 presents results of hypothetically trading \$1 in every stock in our sample every day, using its forecasted daily trading volume. We evaluate in Panel A the OOS mean economic loss (MEL), and Panel B the mean squared error (MSE). For ease of comparison, Panels A′ and B′ normalize these metrics as a percentage loss reduction such that the "oracle" (which represents the perfect prediction $\widehat{v} = \widetilde{v}$) is at 100% accuracy and the baseline ma_5 at 0%. (The percentage reduction in MSE is then the R^2 predicting $\widetilde{\eta}$ as in Table 1 Panel A.) The four columns correspond to different μ values that modulate the economic loss function, representing four different downstream economic tasks relevant to different AUM magnitudes. As the AUM decreases, the relevant economic objective is parameterized by a greater μ, such that overall trading becomes more aggressive. The average trading rate ranges from trading conservatively near the starting position (13% for AUM = \$10b) to trading almost all the way to the target position (95% for AUM = \$10m).

The rows include the statistical prediction methods described in Section 3 (ma_5, ols, nn, and rnn). Additionally, the economic learning approaches (labeled ".econ") conduct fine-tuning on top of the statistical forecasts to optimize the economic loss function under the four different μ's. The lines spanning the columns indicate that the statistical forecasts do not vary with μ, whereas economic learning generates different (z-implied) volume forecasts as μ varies, because μ varies the economic loss from the trade-off between trading cost versus tracking error, based endogenously on the model. The predictor sets include the 8 "tech" signals or "all" of the 175 signals.

Looking first at the statistical performances of the forecasts, model complexity and feature richness reduce MSE. Configuration rnn_{all} yields the highest OOS $R^2 \approx 20\%$ (relative to a ma_5

[32] Some financial machine learning studies, including Jensen et al. (2024), Chen et al. (2023), Cong et al. (2020) and Chen, Pelger, and Zhu (2024) also involve directly training for the economic target.

Table 3: Economic and statistical performance of different methods

	1	2	3	4	1	2	3	4
μ	1.2e-9	6.3e-8	4.7e-7	9.4e-6	1.2e-9	6.3e-8	4.7e-7	9.4e-6
avg $\widetilde{z}$	0.13	0.57	0.78	0.95	0.13	0.57	0.78	0.95
AUM	\$10b	\$1b	\$100m	\$10m	\$10b	\$1b	\$100m	\$10m
A. Mean economic loss (MEL) ($\times 10^{-8}$)					A′. % reduction in mean economic loss			
ma_5	0.1046	3.163	15.41	93.0	0.0	0.0	0.0	0.0
ols_{tech}	0.1041	3.011	14.81	93.2	27.9	29.6	11.1	-0.3
nn_{tech}	0.1043	3.100	14.97	99.7	19.2	12.3	8.0	-12.8
rnn_{tech}	0.1040	2.955	14.37	102.2	33.6	40.8	19.2	-17.7
nn.econ_{tech}	0.1041	2.991	12.35	66.8	31.3	33.6	56.5	50.2
rnn.econ_{tech}	0.1039	2.855	11.78	64.2	39.7	60.4	67.0	55.3
ols_{all}	0.1040	3.024	14.97	94.7	32.4	27.2	8.1	-3.3
nn_{all}	0.1040	3.019	15.07	106.5	33.3	28.2	6.2	-25.9
rnn_{all}	0.1040	3.012	14.78	109.8	34.8	29.5	11.7	-32.1
nn.econ_{all}	0.1039	2.810	11.56	61.9	39.6	69.2	70.9	59.6
rnn.econ_{all}	0.1038	2.812	11.60	66.4	43.7	68.8	70.3	51.0
oracle	0.1029	2.653	9.99	40.8	100	100	100	100
B. Mean squared error (MSE)					B′. R^2 (% reduction in MSE)			
ma_5	——	——	0.437	——	——	——	0.0	——
ols_{tech}	——	——	0.385	——	——	——	12.1	——
nn_{tech}	——	——	0.375	——	——	——	14.3	——
rnn_{tech}	——	——	0.368	——	——	——	15.8	——
nn.econ_{tech}	0.389	0.449	0.457	0.630	11.2	-2.6	-4.5	-44.1
rnn.econ_{tech}	0.392	0.492	0.487	0.481	10.3	-12.4	-11.3	-10.0
ols_{all}	——	——	0.367	——	——	——	16.0	——
nn_{all}	——	——	0.357	——	——	——	18.4	——
rnn_{all}	——	——	0.350	——	——	——	19.9	——
nn.econ_{all}	0.394	0.555	0.590	1.979	10.0	-26.8	-34.9	-352.5
rnn.econ_{all}	0.377	0.440	0.477	0.785	13.9	-0.6	-9.0	-79.5
oracle	——	——	0.00	——	——	——	100	——

Panel A: Mean economic loss (MEL) := avg $loss^{\text{econ}}(\widetilde{v}, \widehat{z}; \mu)$; A′: % reduction in mean economic loss := $(\text{MEL}_{\text{ma5}} - \text{MEL}_m)/(\text{MEL}_{\text{ma5}} - \text{MEL}_{\text{oracle}})$; B: MSE := avg $(\widetilde{v} - \widehat{v}_m)^2$; B′: $R^2 := 1 - \text{avg}\ (\widetilde{v} - \widehat{v}_m)^2/\text{avg}\ (\widetilde{v} - \widehat{v}_{\text{ma5}})^2 = (\text{MSE}_{\text{ma5}} - \text{MSE}_m)/\text{MSE}_{\text{ma5}}$, for each method m and μ ("avg" is the OOS average over i, t). $\widehat{z}$ varies over each method and μ. For statistical methods, $\widehat{v}$ does not depend on μ, hence the horizontal lines indicate the MSE and R^2 do not depend on μ for these methods. These R^2 numbers repeat those from Table 1 Panel A by construction. In the header, the two additional rows help interpret the μ values: avg $\widetilde{z}$ = avg $s(\widetilde{v}; \mu)$ is the average trading rate given true volume; Under each "relevant AUM", the corresponding μ is backed out in portfolio optimization hyperparameter tuning (see Footnote 34 for details on tuning μ).

of volume). However, smaller MSEs do not necessarily lead to better economic performance. For example, nn_{tech} accrues greater MELs than ols_{tech}, albeit nn is more accurate in terms of R^2 (comparing Panels A and B). For the low AUM task in particular, volume forecasts are even worse than

the baseline ma_5 in terms of minimizing economic loss, likely because trading aggressively in this task can be very costly due to overestimating volume. This result is intuitive given the theoretical analysis, and argues why economic fine-tuning produces better OOS portfolio performance.

The economic training methods lead to better economic outcomes. The fine-tuned networks with all the predictors yield the best performance, reaching an OOS economic performance that is about 43% to 70% of the unattainable oracle (perfect forecast) benchmark at various AUM levels. These improved OOS outcomes are not mechanically guaranteed since the fine-tuning is to minimize in-sample loss. The empirical results demonstrate the validity of the economic learning design. Looking at Panel B′, the statistical accuracy retreats after fine-tuning, often to levels even worse than the ma_5 baseline resulting in negative R^2. Hence, to achieve improvements in economic outcomes, the models compromise MSE, and do so by penalizing errors related to overestimating volume more than underestimating volume according to the economic loss function.

Furthermore, for a smaller AUM or, equivalently, a higher μ value, the economic models tend to introduce more statistical bias, as indicated by the increasingly negative R^2 values. This can be explained by the changes in the economic loss functions. With a higher μ, trading is more intensive in general (the curve in Figure 3 shifts to the left), so the penalty for overestimating low volume is more stringent and takes effect earlier (dashed curves in Figure 4 shift to the left). Thus, for smaller AUM, there is a greater difference between the least squares loss and the economic loss.

6 Investment Performance in Trading Experiments

We now apply the analysis to real-world investment portfolios.

6.1 Trading experiment design

We form a set of trades $x_{i,t}$ by applying the portfolio optimization detailed in the previous section to dynamically track a set of given target positions $x^*_{i,t}$. The target positions are not optimized on trading cost considerations, but formed in a separate process focused solely on return prediction or some other objective like hedging or tracking. We evaluate the outcome of the implemented trades

$x_{i,t}$, in the OOS period to see whether a volume prediction method brings improved investment performance net of trading costs and tracking error considerations.

The first experiment mimics a realistic quantitative trading strategy. We simulate an extremely profitable before-cost trading strategy assuming the agent can, with some probability, forecast the realized direction of stock price changes. We experiment with different AUM levels, in which the dollar positions scale linearly while the trading costs increase quadratically. As a result, the optimal μ, which controls the overall trading aggressiveness, varies. The second set of experiments tracks monthly-rebalanced factor portfolios sorted on firm characteristics from the asset pricing literature as the trading targets. These experiments reveal the economic values of volume prediction across the spectrum of investment styles.

Given the target $\{x^*_{i,t}\}$, the implemented trading outcome $\{x_{i,t}\}$ is constructed from $\widehat{z}_{i,t}$, which is formed in the training sample. In particular, $x_{i,t} = x^0_{i,t} + \widehat{z}_{i,t}(x^*_{i,t} - x^0_{i,t})$, where the starting position is formed recursively as $x^0_{i,t} := x_{i,t-1}\left(1 + r_{\mathrm{f},t-1} + \widetilde{r}_{i,t-1}\right)$, which is mechanically adjusted by the arithmetic raw return accrued on day $t-1$.[33] Note the dynamic effect here: the portfolio choice matters for the starting position on the next day, although the optimization does not explicitly consider it. Additionally, the target amount to trade $x^* - x^0$, varies across i, t, which is another aspect abstracted from in the theoretical analysis.

Given the resulting trades $\{x_{i,t}\}$, the investment outcome comes from equation (11).

$$\widetilde{\mathrm{P\&L}}_t = \sum_i x_{i,t}\widetilde{r}_{i,t} - \sum_i TradingCost_{i,t} + Ar_{\mathrm{f},t}.$$

Normalizing by A gives the net-of-cost excess return of the implemented portfolio:

$$\widetilde{r}_{\text{implemented},t} := \frac{\widetilde{\mathrm{P\&L}}_t}{A} - r_{\mathrm{f},t} = \sum_i w_{i,t}\widetilde{r}_{i,t} - A\sum_i \frac{\widetilde{\lambda}_{i,t}}{2}\left(w_{i,t} - w^0_{i,t}\right)^2, \tag{24}$$

where $w_{i,t} := \frac{x_{i,t}}{A}$ and $w^0_{i,t} := \frac{x^0_{i,t}}{A}$ are portfolio weights as a ratio of AUM. We have the familiar result that the before-cost return (the first term) is scale-invariant while the percentage trading

[33] To initiate the recursive calculation, $x^0_{i,t} = 0$ on the first day a stock appears in the sample.

cost due to price impact (the second term) scales with AUM linearly. We report the mean and Sharpe ratio of $\widetilde{r}_{\text{implemented},t}$. Additionally, we also evaluate the annualized turnover as

$$\text{Turnover} := \frac{1}{T}\sum_{i,t}\frac{\left|x_{i,t}-x^0_{i,t}\right|}{2A}\times 252 = \frac{1}{2T}\sum_{i,t}\left|w_{i,t}-w^0_{i,t}\right|\times 252, \tag{25}$$

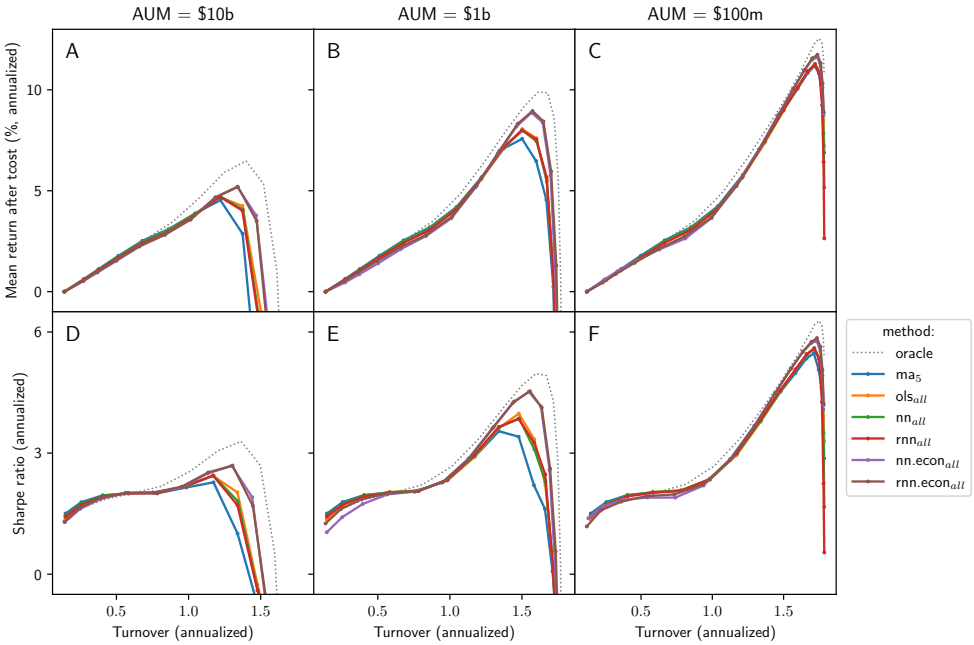
which is scale invariant.

6.2 Implementing a simulated quantitative strategy

We consider a set of trading targets $\{x^*_{i,t}\}$ that simulate a quantitative investment strategy. This strategy is purposely designed to have a very large before-cost Sharpe ratio, but will fail to be profitable after trading costs. The idea is to create a high turnover trading strategy that is only profitable net of costs if trading costs can be managed appropriately. To achieve this goal, we simulate a trading signal that, with 1% chance, perfectly forecasts whether a stock goes up or down over the next five days. The signal is independent across i, t. Following the signal, all stocks are allocated to either the long or the short group. When no signal is received, the stock position stays the same. We let $x^*_{i,t}$ be the equal-weighted long-short strategy in which each leg sums up to 50% of the AUM.

The first row of Table 4 reports the before-cost return and Sharpe ratio of this trading strategy, which is 13.90% with an unrealistically high before-cost Sharpe ratio of 6.97. The second row of Table 4, however, reports the after-cost return and Sharpe ratio of the portfolio taking into account trading costs via price impact, which varies for different fund AUMs. We report results for four AUM levels: \$10b, \$1b, \$100m, and \$10m. Here, we report the after-cost returns and Sharpe ratio for the "all-in" strategy that simply trades every day based on the signal and completely ignores expected volume/liquidity and price impact. For this all-in strategy, at \$10b or \$1b of AUM, the price impact from trading is so large that the strategy loses all of its money (-100% return). That is, trading costs would more than offset any alpha and even wipe out all capital, even on a nearly 7 Sharpe ratio strategy. In this sense, the first experiment is closely related to a lot of quantitative hedge fund strategies that on paper can produce wildly profitable portfolios, but in practice are

Figure 6: Trading experiment performance, with all predictors



Each dot plots the outcome of one trading experiment: the horizontal coordinate is turnover, vertical is the mean return or Sharpe ratio. Each curve is for a method with varying μ values. The predictor set is the full set of predictors for volume.

hindered by implementation costs. In our case, the strategy is rendered completely wealth depleting at AUM of \$10b or \$1b, loses -31.63% at \$100m of AUM, and is only profitable at \$10m AUM, but reduces a 6.97 Sharpe ratio to 2.34 net of trading costs at that size. These numbers of course do not try to reduce trading costs by forecasting volume in order to adjust the trading strategy to account for and mitigate expected trading costs. We turn to this exercise next to see how much volume forecasts can help in making this strategy profitable.

We first evaluate the trading strategy across a grid of μ values (based on different AUMs) that vary the aggressiveness of trades, and then perform the μ-tuning method that allows for the comparison of the highest investment performance attained by each volume prediction method. We run a trading experiment for every volume prediction method and each μ value for various AUM levels. Each experiment plots one dot in Figure 6, with OOS turnover versus net-of-cost mean return (or Sharpe ratio) as the two coordinates. Each curve corresponds to one method, connecting the dots with varying μ values.

To understand the general shape of the curves, first note that under different μ values the

strategies vary from always passively holding ($z = 0$) to trading all the way to the target ($z = 1$). As the implemented portfolio becomes more aggressive (μ increase), the turnover increases. On the vertical axis, the net-of-cost mean return (or Sharpe ratio) first increases as a result of active trading for profit and then bends downward due to trading costs, whose effect shows up with high turnover. The two opposing forces result in hump-shaped curves. For higher AUM, the trading costs' effect is stronger, resulting in lower curves that peak at smaller turnover levels (i.e., lower μ values). In the next subsection, we implement the highest attainable investment outcome by selecting the μ value from the peak of the in-sample curves.

The investment gain of a better trading rate strategy $z(\mathcal{X})$ is shown in the vertical displacement of the curves. The improvement comes from two aspects. First, more closely tracking the target portfolio delivers a higher return. Second, reducing trading costs when liquidity is expected to be low, for a given amount of turnover, lowers the trading cost drag on the portfolio. These two objectives are contradictory in our framework, hence the value of predicting volume more accurately is to strike a better balance in the tradeoff between trading costs and tracking error. At the two extremes of the curves ($\mu = 0$ or $+\infty$), there is no room for tradeoff, and hence predicting volume has no value, in which case all curves converge.

Comparing the methods, we can analyze the sources of improvement. First, using the full set of predictors is better than using only the five-day moving average. Second, holding the information set constant, the neural network model predicts volume better than linear regression. Lastly, given the pre-trained neural network, further fine-tuning the trading strategy by directly optimizing using the economic loss function provides further improvement.

Finally, we tune the hyperparameter μ to report each method's highest attainable investment performance. We choose the μ value that maximizes the in-sample net-of-cost expected return (or Sharpe ratio) and report the out-of-sample performance at the tuned μ. Effectively, μ is selected from the peaks of the in-sample version of the curves (not plotted) and then applied to the OOS curves (as plotted). We expect the in-sample and OOS curves to peak at relatively close μ ranges so that the method attains an OOS performance close to the actual peak of the OOS curves.

[34]The "relevant AUM" row in Table 3 is calculated according to the μ-tuning result under each AUM level under method ma_5.

Table 4: Investment performance in trading experiments

A. Mean return (%, annualized)					B. Sharpe ratio (annualized)			
AUM	\$10b	\$1b	\$100m	\$10m	\$10b	\$1b	\$100m	\$10m
before cost	13.90	13.90	13.90	13.90	6.97	6.97	6.97	6.97
all-in	<-100	<-100	-31.63	9.35	-1.31	-1.27	-0.91	2.34
ma_5	3.88	6.47	11.19	13.20	2.00	2.21	5.47	6.55
ols_{tech}	3.82	7.60	11.28	13.14	2.16	3.32	5.59	6.59
nn_{tech}	3.76	7.30	11.32	13.13	2.14	2.79	5.63	6.59
rnn_{tech}	3.74	7.84	11.33	13.13	2.18	3.58	5.64	6.59
nn.econ_{tech}	4.60	7.20	11.29	13.22	2.13	2.57	5.59	6.63
rnn.econ_{tech}	4.67	8.69	11.59	13.30	2.50	4.32	5.73	6.66
ols_{all}	3.82	7.60	11.28	13.14	2.17	3.35	5.60	6.59
nn_{all}	3.86	7.44	11.28	13.13	2.19	3.09	5.60	6.58
rnn_{all}	3.79	7.55	11.25	13.09	2.18	3.26	5.59	6.56
nn.econ_{all}	4.64	8.87	11.61	13.29	2.18	4.24	5.74	6.66
rnn.econ_{all}	4.68	8.95	11.77	13.30	2.50	4.53	5.85	6.68
oracle	6.47	9.89	12.54	13.56	3.05	4.97	6.28	6.80

For each combination of AUM and method, we report the OOS mean return (Panel A) and Sharpe ratio (Panel B) at the tuned μ, which is selected to maximize the in-sample mean return (or Sharpe ratio) over a grid of μ values.

The results are reported in Table 4. As the third row shows, even if we just use a moving average of five-day volume (ma_5) to forecast tomorrow's volume and alter the aggressiveness of trading, we vastly improve the OOS net-of-cost performance. At an AUM of \$10b, the strategy that uses ma_5 still captures 3.88% returns with a Sharpe ratio of 2.0 net of costs, compared to ignoring volume that wipes out capital due to costs. At \$1b AUM, we capture 6.47% returns with a 2.21 Sharpe ratio, and at \$10m AUM we capture 13.2% net-of-cost returns with a Sharpe of 6.55, which is a huge improvement over the strategy that ignores volume forecasts. Even for a small fund of \$10m, the improvement in net-of-cost performance is quite large.

Applying better prediction methods and using a larger set of predictors improves investment performance uniformly across AUM levels. The economic magnitude of improvement is significant. For \$10 billion of AUM, the average annual return increases from 3.88% when using the baseline prediction method of only a 5-day moving average volume, to 4.68% when making volume predictions with an economic objective optimization embedded within a rnn and using all volume

predictors. For $1 billion AUM, the magnitude of improvement is even greater, going from 6.47% to 8.95%, and more than doubling the Sharpe ratio from 2.21 to 4.53. For smaller AUM ($10m), the improvement is still noticeable but smaller, because price impact shrinks and all methods therefore prescribe trading very aggressively. The investment performance converges to the high before-cost level regardless of the prediction method.[35] The last row of the table reports the results for the oracle or perfect forecast for volume, which represents the maximum benefit the portfolio could receive from knowing tomorrow's volume. The benefits are substantial, particularly for large AUM, and as the table shows, our best prediction method – a rnn with an econ loss function using all predictors – does pretty well at getting close to that (unattainable) maximum. For $10b AUM, our best method delivers a Sharpe ratio of 2.50 compared to 3.05 for the oracle. At $1b AUM, we achieve a Sharpe ratio of 4.53 compared to 4.97 for the oracle, and at $10m AUM, we hit 6.68 and the oracle achieves 6.80 Sharpe. Volume forecasts are much more valuable for high AUM, since price impact is more substantial, and small for low AUM. For a $10b fund, the net-of-cost Sharpe ratio can be improved by 0.50 units or about 25 percent from the baseline ma_5 volume benchmark. That is substantial and on the order of finding new return predictors.[36]

6.3 Implementing factor zoo portfolios

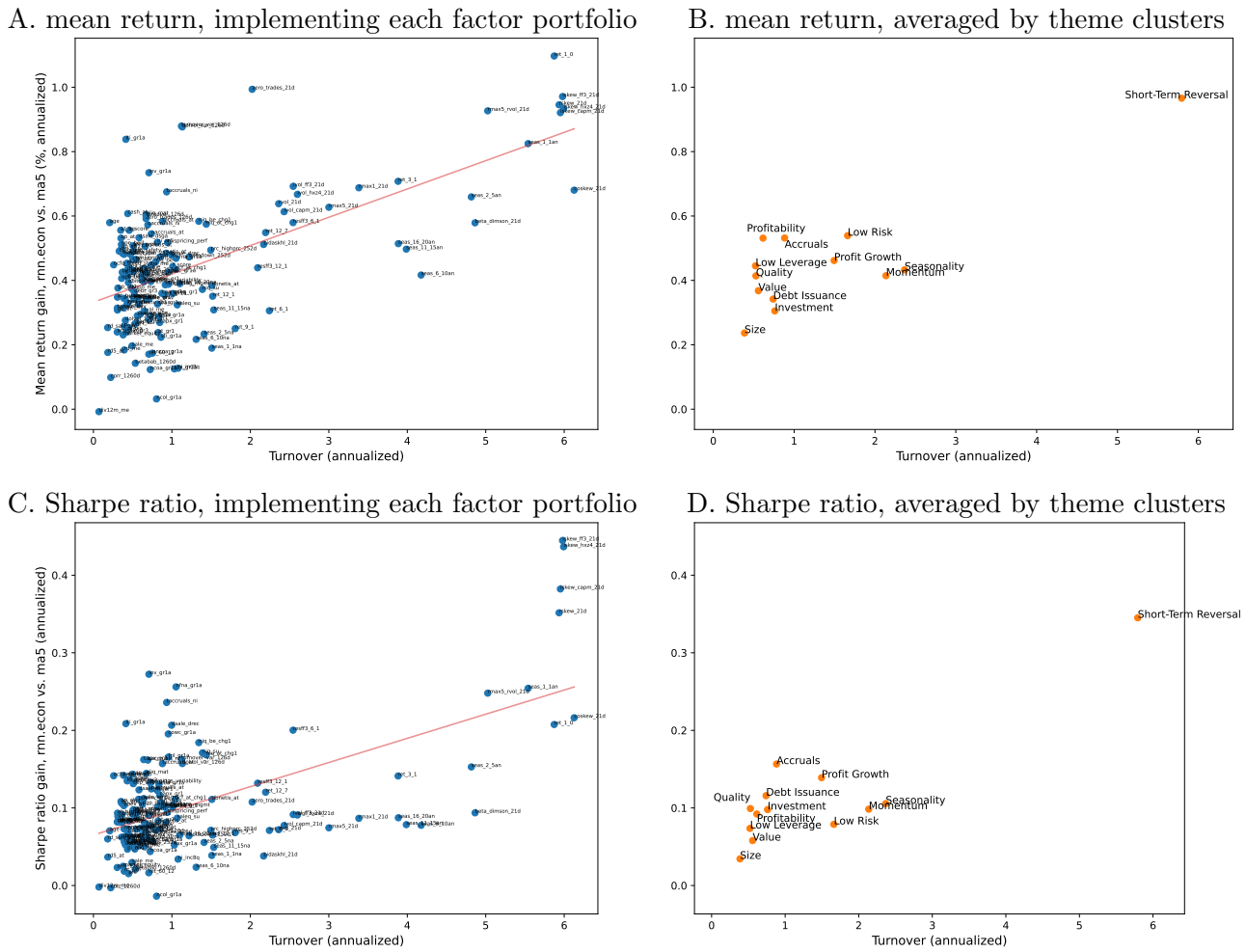
For a second set of trading experiments, we use as trading targets the 153 portfolios sorted on characteristics in the JKP dataset that come from the asset pricing literature. For each of the 153 characteristics, the target $\{x^*_{i,t}\}$ is formed in a standard fashion without considering trading costs. We fix the AUM at $10 billion ($5 billion for each of the long and short legs). The goal here is to examine the improvement in implementation outcomes across different investment styles that pervade the asset pricing literature.

We fix μ across the 153 factors for consistent comparison, and because factor-specific μ tuning is likely to be unstable. For example, consider factors that happen to earn negative realized returns

[35] Future research could consider per-unit trading costs such as bid-ask spread in addition to price impact, which tend to show up as dollar trade sizes shrink.

[36] Appendix B.5 breaks down the mean return and Sharpe ratio improvements reported in Table 4 by firm size groups. We find the gain is robustly positive across all size groups from nano to mega stocks. In particular, the gain is not concentrated in the smaller firms, and larger stocks have greater gains when measured as the percentage relative to the oracle level.

Figure 7: Improvements in implementing each factor portfolio



Each dot implements one JKP factor portfolio. The x-axis is the turnover of the factor portfolio target (i.e., Eq. 25 with $x_{i,t} = x^*_{i,t}, x^0_{i,t} = x^*_{i,t-1}$). Panels A and B show the difference in after-cost mean excess return between implementing with the rnn.econ_{all} and the ma_5. Panels C and D show the corresponding gain in Sharpe ratio. Panels B and D average the points in A and C by style clusters (from JKP).

during the training sample period, the factor-specific optimal μ would be zero – no trading at all. By evaluating at a fixed positive μ, we can address whether the factor that happens to lose money in the sample, would lose less with a better implementation. We pick the μ that optimizes the average gain across all factors, with results robust to perturbations in μ.

Figure 7 plots the gain in mean return when implementing the factor portfolios with the rnn.econ_{all} volume prediction method compared with using the ma_5 volume prediction. The horizontal axis is the turnover of the target factor portfolio. Panel B averages the points by investment

style clusters (from JKP). The plots show volume prediction from the rnn.econ$_{all}$ benefits portfolio implementation across the factors. The average gain in mean after-cost return across factors is 0.44% per year from volume prediction alone using the rnn.econ$_{all}$ model versus the simple moving average. Almost all of the 153 factors have positive gains. With the $10 billion AUM scale, this translates into an additional $44 million per year in implementation cost-saving from improved volume prediction.

Across factors, the gain is larger for those factors with higher turnover. In the right region of Figure 7, some raw factors have a turnover approaching six (600% per year or roughly turning over half of the AUM every month). The gains for these factors, including various short-term reversal strategies, are around 0.5% to 1.0% per year. These factors are constructed with technical signals over a shorter window.[37] In the left part of the figure, even factors with low turnover (those relying on quarterly fundamental signals and signals with greater persistency) show gains that range from 0.2% to 0.6% per year from volume prediction. These improvements are significant.[38] Figure 7, Panels C and D, report gains measured in Sharpe ratio space.

Table 5 reports the net-of-cost performance gains by factor cluster across methods. Short-term reversals have the largest Sharpe ratio improvement (Panel A) from volume prediction, but all factor clusters show Sharpe ratio improvements. The mean return improvements (Panel B) are at least 0.23% to as much as 0.94% per year relative to using the ma$_5$ volume forecast when we use the rnn with an econ loss function and all predictor variables. These mean return improvements (net of costs) are significant and similar in magnitude to return prediction for the set of asset pricing factors the literature has discovered.

[37] Examples: `ret_1_0` short term reversal; `iskew_capm_21d` idiosyncratic skewness from the CAPM; `iskew_ff3_21d` idiosyncratic skewness from the FF3F; `rmax5_rvol_21d` highest 5 days of return; `rskew_21d` return skewness 21d; `seas_1_1an` 1 Year Annual Seasonality; and `coskew_21d` coskewness.

[38] Appendix Figure B.2 plots the same mean return gains in the vertical axis but changes the horizontal axis to the mean return attained by ma$_5$, i.e., the baseline level in the gain calculation. The figure shows that, regardless of the baseline, the gain is independently distributed around a positive center. That is, a better volume prediction is uniformly effective, and the improvement is not concentrated on factors that have positive (or negative) realized returns.

Table 5: Investment performance gain: implementing factor portfolios in each style cluster

	accruals	debt issuance	invest-ment	low leverage	low risk	momen-tum	profit growth	profit-ability	quality	season-ality	short-term reversal	size	value
Panel A: Sharpe ratio gain (annualized)													
ma_5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ols	0.10	0.08	0.07	0.05	0.05	0.07	0.09	0.05	0.07	0.06	0.14	0.04	0.05
rnn	0.07	0.07	0.06	0.04	0.04	0.06	0.08	0.04	0.05	0.04	0.10	0.03	0.03
rnn.econ	0.15	0.11	0.09	0.07	0.07	0.09	0.13	0.08	0.09	0.10	0.32	0.04	0.06
Panel B: mean return gain (percentage points per year)													
ma_5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ols	0.31	0.21	0.22	0.26	0.32	0.31	0.28	0.34	0.28	0.22	0.43	0.21	0.30
rnn	0.25	0.18	0.20	0.21	0.23	0.29	0.24	0.25	0.23	0.16	0.33	0.19	0.21
rnn.econ	0.49	0.32	0.28	0.40	0.49	0.39	0.43	0.49	0.38	0.40	0.94	0.23	0.35

Reported are the after-cost Sharpe ratio gain and mean excess return gain between implementing the factor portfolios with different volume prediction models versus just using ma_5 for volume. Results are reported by factor/style clusters (from JKP).

6.4 Applications, assumptions, and economic importance

While we run several trading experiments that 1) mimic a quantitative hedge fund strategy and 2) focus on academic asset pricing factors from the literature, our framework and prediction methods could be applied to a variety of tasks. The tradeoff between implementation costs and tracking error can be applied to hedging, benchmarking, satisfying investment constraints or other objectives (e.g., ESG). The economic benefit of predicting volume will vary by application and by the assumptions we make in our framework. For our experiments of quant trading strategies and asset pricing factors, we find substantial benefits to forecasting volume based on our modeling assumptions. It is useful, however, to consider whether our assumptions over- or under-estimate these benefits and what the likely impact of modifying or extending the model might be.

Our model is admittedly and purposely simple, so as to showcase the benefits of volume prediction in an intuitive way that future research can extend. Despite this, we still find large benefits. However, we consider here how different assumptions might impact those benefits. First, we only predict daily trading volume, but a richer model could predict intra-daily volume, signed intra-daily volume, and try to separate informed from uninformed volume. Presumably, these more nuanced measures of volume would have even more value than total volume, as they relate more strongly to price impact costs. Second, we use a linear price impact function from Kyle's (1985) model, but empirical estimates of this function suggest it is a power function with a coefficient < 0.5 (Frazzini, Israel, and Moskowitz, 2018). Such functions would reduce the benefit of volume prediction, espe-

cially at large AUM size. We also only estimated volume using the most recent five years of data, where trading costs have decreased significantly over time. Using a longer time series of data would almost surely increase the estimated benefit of volume prediction, as price impact is much larger historically (Frazzini, Israel, and Moskowitz, 2018). We also estimate Kyle's λ using the Frazzini, Israel, and Moskowitz (2018) results, which is much smaller than λ estimated by TAQ data in the literature. The latter would give a substantively larger price impact and assign even more value to volume prediction.

In addition, a more exhaustive search for prediction variables and models that forecast volume more accurately could translate into even larger economic benefits. For example, Frazzini, Israel, and Moskowitz (2018) show that market and idiosyncratic volatility are significant variables that impact real-world price impact costs. Using similar techniques as shown here to forecast market and individual stock volatility could further improve net-of-cost performance. While our objective is to focus on non-return variables to illustrate the power of using an economic objective for the prediction problem in a transparent and clear way, incorporating multiple variables in a more sophisticated trading cost model is a natural next step. Focusing on volatility, for instance, this would not only entail two prediction problems, but would add additional complications because volume and volatility are correlated and in different ways – being negatively correlated in the time series and positively correlated in the cross-section. Hence, it is difficult to separate the two effects and assess the benefits of one versus the other. Therefore, whether the benefits of volume prediction would be mitigated or exaggerated by including volatility prediction simultaneously is not clear. The impact would also likely vary significantly across fund sizes. Some other promising candidates for additional features of a trading cost model are lead-lag volume relations across stocks, more seasonal indicators, and other market microstructure variables. Our goal, however, is to simply translate the prediction problem into economic consequences, which yield interesting insights, and our simple framework provides that, and can be extended.

Finally, we separate the volume prediction problem from the return moment modeling problem. Combining both could generate further portfolio improvements and the interaction between them could be enlightening. Of course, attempting all of this at once obscures intuition for what, specif-

ically, is leading to improved OOS performance. The current paper, by focusing just on trading costs, and abstracting from predictability of return moments, provides clear intuition and showcases the potential value of non-return predictability for asset pricing. The next steps would be to incorporate these insights with return prediction from an already extensive literature.

7 Conclusion

We translate a prediction problem into an economic problem to quantify the economic value of forecasting trading volume for asset pricing. We convert volume forecast errors into net-of-cost portfolio performance through a simple, but intuitive framework. A key insight is that using the economic loss function, in lieu of a statistical loss function, produces more valuable volume forecasts that lead to better portfolio and trading decisions. This insight highlights the tradeoff between statistical efficiency and economic decision-making, where volume is highly predictable, but its impact on portfolio performance is non-linear and asymmetric. While the value of volume prediction depends on the assumptions and applications we consider, we find significant value using a simple and conservative model for several meaningful investment applications. For example, applied to asset pricing portfolios used in practice and using an economic loss function with machine learning, large data signals, and exploiting the virtue of complexity, we find that the benefits of volume prediction can be as valuable as return prediction.

Trading volume prediction is an interesting and understudied research area worthy of further exploration. Our approach of using an economic model to guide the forecasting problem may be useful in furthering our understanding of trading volume for asset pricing. More generally, a framework similar to ours may be appealing for predicting other non-return variables for asset pricing that potentially opens up a host of new pursuits in this literature.

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Internet Appendix

Trading Volume Alpha

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A Technical details

A.1 Neural network implementation details

The nn architecture consists of three fully-connected hidden layers with 32, 16, and 8 neurons, respectively.

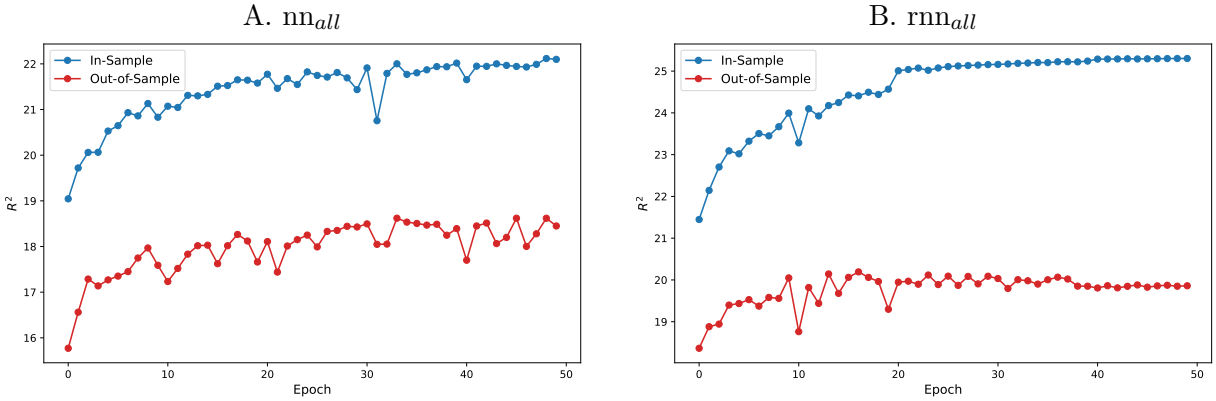
The rnn architecture is similar to that of the nn. The first (bottom) hidden layer in the 3-layer network is upgraded to an lstm layer with 32 hidden states and cell states, respectively. The remaining two layers are unchanged: fully connected with 16 and 8 neurons, respectively.

The formulas for the number of parameters in nn and rnn, as reported in Table 1 B, are stated below. For nn (three fully-connected hidden layers with 32-16-8 neurons), the formula is $(\#\text{ of predictors} + 1) \times 32 + (32 + 1) \times 16 + (16 + 1) \times 8 + (8 + 1)$. In rnn, the first hidden layer has 32 hidden states and 32 cell states with four gates, changing the formula to $(\#\text{ of predictors} + 32 + 1) \times 32 \times 4 + (32 + 1) \times 16 + (16 + 1) \times 8 + (8 + 1)$.

In training the rnn, we implement the "many-to-one" type data pipeline, where the model recursively processes a sequence of 10 inputs, $X_{i,t-9}, \ldots, X_{i,t}$, and produces a single output $\widehat{\eta}_{i,t}$ to calculate the training loss at each data point $\{i, t\}$. (When increasing the sequence length from 10 to 50, the results had minimal improvements but required much larger GPU memory and longer training time.) For data points at the beginning of a stock's observed period where lagged predictors (e.g., $X_{i,t-9}$) are not available, we fill in with zero vectors.

In training both nn and rnn models, we use the Adam optimizer, with default learning rates and other parameters. The batch size is 1024. The $\widetilde{\eta}$ prediction models are trained with 50 epochs. For the sake of clear benchmarking, we do not adopt early stopping, weight dropout, or hyperparameter tuning with cross-validation, though these techniques could further boost the prediction accuracy. The machine learning program is implemented with the PyTorch package.

Figure A.1: Learning curves



R^2 of the $\widetilde{\eta}$ prediction models as training progresses (epochs).

The learning curves show the gap between the IS and OOS R^2 is relatively small, and does not widen with continued training. This indicates limited in-sample overfitting at this neural network configuration. The rnn learning curves show slightly more severe overfitting, though the OOS learning curve is still relatively stable as training continues. The learning curves display fluctuation due to the randomness of the stochastic gradient descent, though the extent to which is not severe. We also note the exact results depend on the inherent randomness of the training program. We find the quantitative results are insensitive to random seeds and report the average of five independent runs to obtain a robust evaluation outcome.

Table A.1 summarizes computational costs in terms of the training times and memory usage for nn and rnn with all predictors for a single random seed. These experiments were conducted on a system equipped with an Nvidia A100 GPU with 40 GB of GPU memory, an AMD EPYC 7713 64-Core Processor @ 1.80GHz with 128 cores, and 1.0TB of RAM, running Ubuntu 20.04.4 LTS.

Table A.1: Training time and memory usage

	Training time (hours)	CPU memory usage (GB)	GPU memory usage (GB)
nn_{all}	0.48	10.87	1.22
rnn_{all}	0.63	144.98	1.48

B Additional empirical results

B.1 Prediction accuracy result in a different R^2 measure

Table B.2 recasts the R^2's in Table 1 in terms of the explained percentage of the total variation of $\widetilde{v}$. There is no change in the model, fitted values, nor the mean squared error. Only the R^2 is calculated with a different numerator. Under this metric, the R^2's are always high since the benchmark moving average explains $\widetilde{v}$ to a large extent already (93.68%), while machine learning and more predictors still provide accuracy improvement.

Table B.2: Table 1 prediction accuracy result in a different R^2 measure

cumulatively adding predictor sets total number of predictors	tech 8	fund-1 14	fund-2 161	calendar 165	earnings 175
A: R^2 relative to $\widetilde{\eta}$ (log dollar volume shock), Table 1A repeated for reference					
ma_5	0				
ols	12.09	12.26	12.27	14.85	15.99
nn	14.31	14.90	14.42	17.13	18.45
rnn	15.80	16.25	15.47	18.12	19.86
A': R^2 relative to $\widetilde{v}$ (log dollar volume)					
ma_5	93.68				
ols	94.44	94.45	94.45	94.62	94.69
nn	94.58	94.62	94.59	94.76	94.85
rnn	94.68	94.69	94.64	94.86	94.93

Panels A and A' respectively express the OOS prediction accuracy in two different R^2's. The R^2 is calculated with ma_5 as the benchmark: $R^2 = 1 - \text{MSE}/\text{avg}(\widetilde{v} - \text{ma}_5)^2$. Panel A' changes the benchmark to "predicting 0" essentially: $R^2 = 1 - \text{MSE}/\text{avg}(\widetilde{v} - \text{avg}(\widetilde{v}))^2$, where $\text{MSE} := \text{avg}(\widetilde{v} - \widehat{v})^2 = \text{avg}(\widetilde{\eta} - \widehat{\eta})^2$ and $\text{avg} := \frac{1}{|\text{OOS}|}\sum_{i,t\in\text{OOS}}$ is the OOS average.

B.2 Volume forecasting by standardizing target variable $\widetilde{\eta}$

In Section 3, we forecast volume by having the supervision target variable as the log dollar volume *shock* defined as daily log dollar volume minus the moving average in the past five days, $\widetilde{\eta}_{i,t} := \widetilde{v}_{i,t} - [\text{ma}_5]_{i,t}$, where $[\text{ma}_5]_{i,t} := \frac{1}{5}\left(\widetilde{v}_{i,t-1} + \cdots + \widetilde{v}_{i,t-5}\right)$. We have explained that predicting the log dollar volume shock is essentially predicting the log dollar volume itself, which is simply the sum of the shock forecast and the moving average: $\widehat{v}_{i,t} = \widehat{\eta}_{i,t} + [\text{ma}_5]_{i,t}$.

In this Appendix, we consider a related but different method by further standardizing the target variable $\widetilde{\eta}$, and having it as the supervision target of the prediction models. The idea is that the standardized target might be more well-behaved and be more amenable for the neural networks to form accurate predictions. However, we find that this standardization method does not improve the prediction accuracy compared to the benchmark method.

The procedure is the following. We standardize the target variable $\widetilde{\eta}$ by dividing its rolling standard deviation over the past 22 days: $\widetilde{\zeta}_{i,t} := \frac{\widetilde{\eta}_{i,t}}{[\text{stddev}]_{i,t}}$, where $[\text{stddev}]_{i,t} := \sqrt{\frac{1}{22}\left(\widetilde{\eta}^2_{i,t-1} + \cdots + \widetilde{\eta}^2_{i,t-22}\right)}$.[39] Then we run the same ols, nn, and rnn models to predict $\widetilde{\zeta}$ with the same sets of predictors. We back out the forecasted values by reversing the standardization: $\widehat{\eta}_{i,t} = \widehat{\zeta}_{i,t} \times [\text{stddev}]_{i,t}$. And finally, we report the same OOS R^2 evaluations as in Section 3.

The results are reported in Table B.3, and are compared with those of the benchmark method in Table 1 for ease of comparison. We find that although the standardization method still yields positive R^2 values for the nn and rnn methods, these are consistently lower than the benchmark values reported in Table 1. The relative comparisons between the nn and rnn methods as well as among the different sets of predictors are still the same as the benchmark method, highlighting the robustness of the results presented in the main text.

B.3 Prediction results in firm size groups and "mixture of experts" forecasts

Table B.4 Panel A provides additional assessments of prediction accuracy by evaluating volume forecasts in different size groups. We use the five groups from the JKP data sorted on the firms'

[39] We have experimented with five-day standardization as well and found the results are similar.

Table B.3: Prediction accuracy (R^2 in %) using the standardization method

cumulatively adding predictor sets	tech	fund-1	fund-2	calendar	earnings
A: R^2 original method (repeating Table 1 Panel A)					
ma_5	0				
ols	12.09	12.26	12.27	14.85	15.99
nn	14.31	14.90	14.42	17.13	18.45
rnn	15.80	16.25	15.47	18.12	19.86
B. R^2 using the standardization method					
ols	-0.06	-0.06	-0.07	-0.06	-0.04
nn	11.79	12.05	11.30	12.29	13.73
rnn	14.42	14.51	12.36	14.94	16.23

market capitalization.[40]

Table B.4: Prediction accuracy (R^2 in %) in different size groups and "mixture of experts"

size group	jointly	nano	micro	small	large	mega
training obs	2,522,619	300,790	797,880	680,209	479,839	263,901
testing obs	1,893,067	273,792	467,413	552,503	384,819	214,540
A: pooled training evaluated in size groups and jointly (same models as in Table 1)						
ols_{all}	15.99	13.32	12.60	20.90	25.49	26.16
nn_{all}	18.45	15.80	14.86	23.71	27.76	29.12
rnn_{all}	19.86	16.63	16.14	26.00	30.50	32.02
B: size group training evaluated in size groups and jointly (mixture of experts)						
$ols+moe_{all}$	16.34	13.68	12.73	21.43	25.93	27.47
$nn+moe_{all}$	17.78	15.29	14.43	22.69	26.57	27.71
$rnn+moe_{all}$	18.26	15.24	14.71	24.76	29.02	30.99

Panel A evaluates the benchmark models (pooled training) in the five size groups, respectively, in the OOS period. Each model uses "all" 175 predictors. Column "jointly" repeats Table 1, Panel A, last column. Panel B trains "expert" models for each size group separately and evaluates them in their corresponding size groups in the OOS period. Column "jointly" evaluates the mixture of experts (moe) model, which predicts with the corresponding expert model trained on the same size group for each OOS data point. Each R^2 value is the average of five runs, same as Table 1.

The prediction accuracy increases as firm size increases, regardless of the prediction method. The R^2's evaluated in the mega firms are roughly twice those of the nano firms. As explained in the main text, smaller firms have a greater magnitude of unexpected trading volume shocks that

[40]The five groups are defined according to the market capitalization breakpoints of NYSE stock percentiles: mega stocks, greater than the 80th percentile; large, 50–80; small, 20–50; micro, 1-20; and nano, below the 1st percentile.

are harder to predict. This result makes sense since small firms are volatile and have low trading volume, hence unexpected events that give rise to volume spikes are more likely for these firms. This finding also indicates that in addition to small firms being less liquid on average, their liquidity is also less predictable and more volatile. Hence, tiny firms are not only costly to trade in general, but their costs are less predictable. These results are intuitive and suggest that our prediction models are capturing true variation in volume and not simply noise.

We also examine whether firms of different size groups should be modeled differently. We train a model on each size group separately to attempt to better capture the heterogeneity across the firm size dimension rather than pooling all firms in the same model. We implement a simple mixture of experts (moe) method, where each size group is trained separately to form "expert" models and then compared against the pooled training model in Panel A.

The moe improves performance of the ols method. Comparing the first lines in Panels A and B, linear models catered to different size groups are more accurate than the pooled ols. For nn and rnn, however, the sample size reduction outweighs the potential benefits of separate training, making the mixture of expert models less accurate (either conditioning on size groups or jointly). Since the potential non-linear effects of firm size are already allowed in the neural networks, forcing size groups into different models is less effective. For this reason, we stick with pooled training samples.

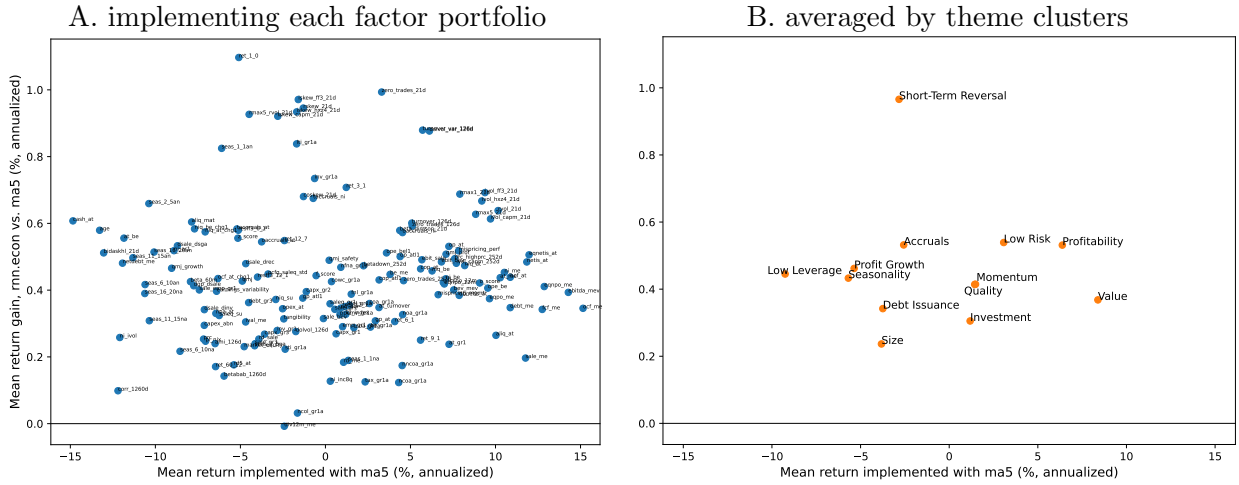
B.4 Additional results of implementing factor zoo portfolios

We provide results on the trading experiments that implement the factor zoo portfolios in addition to Subsection 6.3.

Figure B.2 has the same vertical axis as Figure 7, which is the improvement in after-cost mean excess return from ma_5 to rnn.econ_{all}. The horizontal axis changes to the mean excess return achieved with the ma_5 method, i.e., the baseline level of the improvement. The plot shows the gain is distributed around a positive center uncorrelated with the baseline. That is, a better volume prediction is uniformly effective, and the improvement is not concentrated on factors that have positive (or negative) realized returns.

Figure 7, Panels C and D, show the Sharpe ratio version of Figure 7, Panels A and B, by reporting the gain in Sharpe ratio instead of the mean return. The plot shows a similar pattern to the one reported in the main text. The gains in Sharpe ratios are larger for those factors with higher turnover, reaching around 0.3 to 0.4 per year.

Figure B.2: Mean return improvements in implementing each factor portfolio



The same plot as Figure 7, but changing the x-axis to mean return achieved with the ma_5 method, i.e., the baseline of the gain.

B.5 Volume prediction's economic values in different firm size groups

In Section 6, we show volume prediction has significant economic value in portfolio tracking tasks. We now further evaluate the economic value in different firm size groups. We break down the improvement of the portfolio tracking experiment in mean return and Sharpe ratio by firm size groups as defined in JKP. We find the gain is robustly positive across all size groups. In particular, the gain is not concentrated in the smaller firms as the mega and large firms also have significant mean return and Sharpe ratio gains. The ratio of the gain relative to what the oracle model (which represents the impossible upper bound) could have achieved is monotonically higher in the larger size group, whereas the absolute level of the gain varies.

The detailed evaluation method is the following. We do not retrain the volume prediction models nor create new portfolio implementation tasks. Instead, we take the same portfolio trading outcome as reported in Section 6.2 and break down the sample panel of firm-day portfolio outcomes

$\{x_{i,t}\}$ by the five size groups. This is as if our portfolio tracking task is conducted jointly by five trading desks, each specialized in the stocks of a size group. We respectively evaluate each desk's performances (mean return and Sharpe ratio). We focus on the simulated tracking task with a $10 billion AUM as defined in Section 6.2.

Table B.5: Volume "alpha" (improvement of investment performance) in different size groups

A. Mean return (%, annualized)						B. Sharpe Ratio (annualized)				
	mega	large	small	micro	nano	mega	large	small	micro	nano
ma_5	5.54	6.18	3.58	2.58	2.54	1.81	2.24	1.36	0.86	0.46
rnn.econ_{all}	6.09	7.46	4.51	2.93	3.46	2.55	3.46	1.68	1.15	0.54
gain	0.55	1.28	0.93	0.35	0.92	0.74	1.22	0.32	0.29	0.08
oracle	6.28	8.27	5.98	4.26	9.84	2.60	3.65	1.94	1.40	1.22
pct gain to oracle	75	61	39	21	13	94	87	55	54	11

Line "gain" is the difference between the first two rows in each column (rnn.econ_{all} - ma_5). Line "pct gain to oracle" is the percentage of the gain of the rnn.econ_{all} relative to what the oracle model could have achieved over the ma_5 strategy. That is, "pct gain to oracle" = (rnn.econ_{all} - ma_5) / (oracle - ma_5)×100.

We find the trading volume alpha, i.e. the improvement in investment performance brought by volume prediction, is robustly positive across all size groups. The mega (NYSE 80th market cap percentile and above) and large (50-80 percentiles) firms have significant mean return and Sharpe ratio gains. We also report what the oracle model could have achieved, which is the highest portfolio outcome should the true volume be revealed to the investor. There is a varying level of oracle performance across size groups, which has nothing to do with the volume prediction but reflects the tracking target's different performance in different size groups as well as different levels of liquidity across size groups. To control for that variation, we calculate the percentage of the gain relative to the oracle performance. We find the percentage gain is monotonically higher in a larger size group. We believe this is because larger firms tend to have higher volume levels and overall more aggressive trading, which are most susceptible to highlighted trading costs when volume is over-predicted; and our trading intensity strategy (z) with economic learning is particularly effective in these cases to avoid the high costs of trading too much when volume is low.

C Additional theoretical analysis

C.1 Microfoundation of the tracking error penalty in the portfolio objective

The tracking error penalty term in the portfolio optimization objective function can be economically founded. We show the quadratic tracking error penalty can be derived from a mean-variance utility function. This analysis connects the target positions x_i^* to the before-cost mean-variance efficient portfolio weight, which increases in the asset's return expectations as well as the total portfolio size (AUM). It also provides a microfoundation of the hyperparameter μ and the quadratic penalty: as the position deviates away from the target, the mean-variance loss increases in a quadratic fashion.

Assume the agent has a mean-variance utility function, $U = \mathbb{E}A' - \frac{\gamma}{2A}\mathbb{V}\text{ar}A' - TCost$, where A' is the before-cost investment outcome, A is the initial wealth (AUM), and γ is the risk aversion coefficient, $TCost$ is the term for the objective of minimizing transaction costs (not the focus here as we are justifying the tracking error part). A' is the outcome of the portfolio strategy: $A' = A(1 + r_\text{f}) + \sum_i x_i r_i$, where x_i is the dollar position in risky asset i with excess return r_i. Assume $\mathbb{E}r_i = m_i$, $\mathbb{V}\text{ar}r_i = \sigma^2$, and zero covariances. The agent's portfolio optimization problem is choosing $\{x_i\}$ to maximize U.

Then, the objective function is

$$U = A(1 + r_\text{f}) + \sum_i x_i m_i - \frac{\gamma}{2A}\sum_i x_i^2\sigma^2 - TCost \tag{26}$$

$$= \sum_i \left(-\frac{\gamma\sigma^2}{2A}x_i^2 + m_i x_i\right) + A(1 + r_\text{f}) - TCost \tag{27}$$

$$= -\frac{\gamma\sigma^2}{2A}\sum_i \left(x_i^2 - \frac{2A}{\gamma\sigma^2}m_i x_i\right) + A(1 + r_\text{f}) - TCost \tag{28}$$

$$= -\frac{\gamma\sigma^2}{2A}\sum_i \left(x_i - \frac{A}{\gamma\sigma^2}m_i\right)^2 + \left(\frac{A}{2\gamma\sigma^2}\sum_i m_i^2 + A(1 + r_\text{f})\right) - TCost \tag{29}$$

The first term matches the tracking error term modeled in Eq. 14. The second term is the before TCost utility at the zero tracking error portfolio ($x = x^*$). This term is constant of the x choice, hence can be ignored in the optimization problem.

Comparing the first term with the tracking error modeled in Eq. 14, we see the target portfolio

is $x_i^* = \frac{A}{\gamma\sigma^2} m_i$. As expected, the target positions are the result of Markowitz mean-variance optimization. They are proportional to the return expectations in the cross section. They scale linearly with the AUM (A) and inversely with the risk aversion coefficient and volatility. Additionally, the overall tracking error penalizing coefficient $\mu = \frac{\gamma\sigma^2}{A}$, which is decreasing in AUM. The rationale is that the quadratic penalty stems from the quadratic risk change as the position deviates away from the target, and that the absolute risk aversion coefficient decreases with the wealth level. Although the main analysis takes μ as a hyperparameter ignoring its microfoundation, we still observe the negative relationship between the tuned μ and the economically relevant AUM (e.g., in Section 5.3).

C.2 The economic task as predicting $\widetilde{z}$

We have shown the economic task of choosing the z strategy, $\min_{z(\cdot)} \sum_{i,t\in\text{train}} loss^{\text{econ}}(\widetilde{v}_{i,t}, z(\mathcal{X}_{i,t}); \mu)$, can be seen as a prediction task for $\widetilde{v}$ with the economic loss function: $\min_{v(\cdot)} \sum_{i,t\in\text{train}} loss^{\text{econ}}_{\text{vv}}(\widetilde{v}_{i,t}, v(\mathcal{X}_{i,t}); \mu)$. In this appendix, we provide the equivalent representation as a prediction problem of the oracle trading rate $\widetilde{z} := s(\widetilde{v}; \mu)$.

Define $loss^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu) := loss^{\text{econ}}(s^{-1}(\widetilde{z}; \mu), z; \mu)$, where $s^{-1}(\ \cdot\ ; \mu)$ is the inverse of $s(\ \cdot\ ; \mu)$ function. Under this definition, the economic task can be viewed as the problem of looking for a function $z(\cdot)$ that maps $\mathcal{X}$ into z to minimize the training sample average loss:

$$\min_{z(\cdot)} \sum_{i,t\in\text{train}} loss^{\text{econ}}_{\text{zz}}(\widetilde{z}_{i,t}, z(\mathcal{X}_{i,t}); \mu) \tag{30}$$

According to the definition, the analytical expression of $loss^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu)$ is

$$loss^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu) = \frac{\mu}{\widetilde{z}}(z - \widetilde{z})^2 + \mu(1 - \widetilde{z}). \tag{31}$$

In this expression, the loss can be seen as the squared z prediction error weighted by $\frac{\mu}{\widetilde{z}}$. The last term is constant of choice z so can be ignored in optimization. It equals $loss^{\text{econ}}_{\text{zz}}(\widetilde{z}, \widetilde{z}; \mu)$, the baseline loss incurred even with the perfect prediction.

To derive this expression, notice $\widetilde{z}$ is already defined as the perfect trading given $\widetilde{\lambda}$ or $\widetilde{v}$.

$$\widetilde{z} = \frac{\mu}{\mu + \widetilde{\lambda}} \Longrightarrow \widetilde{\lambda} = \frac{\mu}{\widetilde{z}} - \mu \tag{32}$$

Then, start from Eq. 18, represent $\widetilde{\lambda}$ with $\widetilde{z}$ and then complete the square:

$$\begin{aligned} loss^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu) &= \widetilde{\lambda} z^2 + \mu(1-z)^2 \\ &= \left(\frac{\mu}{\widetilde{z}} - \mu\right) z^2 + \mu(1-z)^2 \\ &= \frac{\mu}{\widetilde{z}}(z - \widetilde{z})^2 + \mu(1 - \widetilde{z}) \end{aligned}$$

We show $loss^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu)$ is still meaningfully different from the standard squared error loss function, and that $\mathbb{E}\left[\widetilde{z}|\mathcal{X}\right]$ will not be the optimal choice either further below.

C.3 Economic loss functions are not in Bregman class

We show functions $loss^{\text{econ}}_{\text{vv}}(\widetilde{v}, v; \mu)$ and $loss^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu)$ are not in the Bregman class, for all μ.

Without loss of generality, any loss function $F(p, q)$ can be normalized as $\bar{F}(p, q) := F(p, q) - F(p, p)$, such that the function acquires the convenient property that $\bar{F}(p, p) = 0$, and that the solution to the optimization problem $\min_{q \in \sigma\{\mathcal{X}\}} \mathbb{E}\left[F(p, q)\right]$ does not change. Therefore, in the following propositions, we normalize accordingly and consider $\overline{loss}^{\text{econ}}_{\text{vv}}(\widetilde{v}, v; \mu) := loss^{\text{econ}}_{\text{vv}}(\widetilde{v}, v; \mu) - loss^{\text{econ}}_{\text{vv}}(\widetilde{v}, \widetilde{v}; \mu)$ and $\overline{loss}^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu) := loss^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu) - loss^{\text{econ}}_{\text{zz}}(\widetilde{z}, \widetilde{z}; \mu)$.

The definition of the Bregman loss function is (Banerjee, Guo, and Wang, 2005):

Definition 1. *Let $\phi : \boldsymbol{R}^d \to \boldsymbol{R}$ be a strictly convex differentiable function, then, the Bregman loss function $D_\phi : \boldsymbol{R}^d \times \boldsymbol{R}^d \to \boldsymbol{R}$ is defined as:*

$$D_\phi(p, q) := \phi(p) - \phi(q) - \langle p - q, \nabla\phi(q) \rangle \tag{33}$$

We consider the simpler case where p, q are scalars. In this case, a Bregman function has the property that its partial second derivative in the first argument is independent of the second

argument.

$$\frac{\partial^2 D_\phi(p,q)}{\partial p^2} = \phi''(p) \tag{34}$$

The propositions below rely on this property.

Proposition 2. *Function* $\overline{loss}^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu)$ *is not in the Bregman class, for all* μ*.*

Proof. We verify that $\overline{loss}^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu)$ violates the property in Eq. 34.

$$\overline{loss}^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu) = \frac{\mu}{\widetilde{z}}(z - \widetilde{z})^2 \tag{35}$$

$$\frac{\partial^2 \overline{loss}^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu)}{\partial \widetilde{z}^2} = \frac{2\mu z^2}{\widetilde{z}^3} \tag{36}$$

It is clear that this is not irrelevant to z. □

Proposition 3. *Function* $\overline{loss}^{\text{econ}}_{\text{vv}}(\widetilde{v}, v; \mu)$ *is not in the Bregman class, for all* μ*.*

Proof. Given the property in Eq. 34, a Bregman loss function must be unbounded as $p \to +\infty$. This is because for any fixed q, when $p > q$, $D_\phi(p, q)$ is increasing and convex in p.

However, we verify that $loss^{\text{econ}}_{\text{vv}}(\widetilde{v}, v; \mu)$ is bounded by showing it converges to a finite number as $\widetilde{v} \to +\infty$.

$$\begin{aligned}
\overline{loss}^{\text{econ}}_{\text{vv}}(\widetilde{v}, v; \mu) := & \frac{0.2\exp(-\widetilde{v}) + \mu\left(\exp\left(-v + \log 0.2 - \log\mu\right)\right)^2}{\left(1 + \exp\left(-v + \log 0.2 - \log\mu\right)\right)^2} \\
& - \frac{0.2\exp(-\widetilde{v}) + \mu\left(\exp\left(-\widetilde{v} + \log 0.2 - \log\mu\right)\right)^2}{\left(1 + \exp\left(-\widetilde{v} + \log 0.2 - \log\mu\right)\right)^2} \\
= & \frac{(0.2\mu)^2 \exp(-\widetilde{v})\left(\exp(\widetilde{v}) - \exp(v)\right)^2}{\left(\mu\exp(\widetilde{v}) + 0.2\right)\left(\mu\exp(v) + 0.2\right)^2}
\end{aligned} \tag{37}$$

By L'Hôpital's rule, we have the limit of it as:

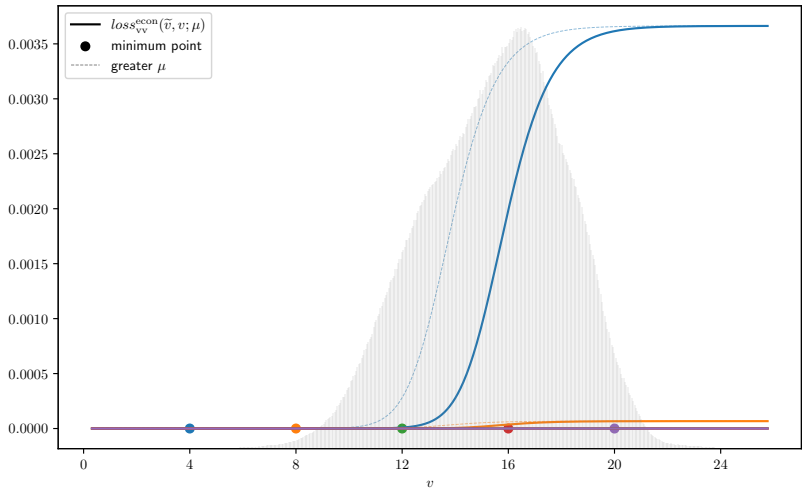
$$\begin{aligned}
\lim_{\widetilde{v}\to\infty} \overline{loss}^{\text{econ}}_{\text{vv}}(\widetilde{v}, v; \mu) &= \lim_{\widetilde{v}\to\infty} \frac{(0.2\mu)^2\left(\exp(\widetilde{v}) - \exp(2v - \widetilde{v})\right)}{\mu\left(\mu\exp(v) + 0.2\right)^2\exp(\widetilde{v})} \\
&= \frac{0.04\mu}{\left(\mu\exp(v) + 0.2\right)^2}
\end{aligned} \tag{38}$$

□

C.4 Further analysis on the loss functions

The following is a visualization of the economic loss function in addition to the one in Figure 4 Panel B. The vertical axis is changed to the linear scale from the log scale. For large $\widetilde{v}$ (12, 16, 20, in green, red, purple), the curves are indistinguishable from a flat line because the blue curve is at a much greater magnitude, which is for the loss of overestimating low actual volume ($\widetilde{v} = 4$).

Figure C.3: Economic loss function in linear scale



Note: same as Figure 4 Panel B but with the vertical axis in linear scale.

The following proposition formally states the asymmetric property of the loss function for over/under-estimating volume.

Proposition 4. *Consider two symmetrical cases with low and high liquidity* $\widetilde{v}_1$ *and* $\widetilde{v}_2$ *such that* $\widetilde{z}_1 = 1 - \widetilde{z}_2 < 0.5$. *Suppose one makes an overestimation in the low-volume case* $\widehat{v}_1 = \widetilde{v}_1 + \varepsilon$, *comparing with an equal amount of underestimation in the high-volume case* $\widehat{v}_2 = \widetilde{v}_2 - \varepsilon$, *the additional loss incurred in the first case is greater than the second:*

$$loss_{vv}^{econ}(\widetilde{v}_1, \widehat{v}_1; \mu) - loss_{vv}^{econ}(\widetilde{v}_1, \widetilde{v}_1; \mu) > loss_{vv}^{econ}(\widetilde{v}_2, \widehat{v}_2; \mu) - loss_{vv}^{econ}(\widetilde{v}_2, \widetilde{v}_2; \mu), \quad \forall \mu > 0. \tag{39}$$

Proof. We first show that, given $\widetilde{z}_2 = 1 - \widetilde{z}_1$, as well as $\widehat{v}_1 - \widetilde{v}_1 = \widetilde{v}_2 - \widehat{v}_2 = \varepsilon$, we have $\widehat{z}_2 = 1 - \widehat{z}_1$,

and that $\widetilde{z}_1 - \widehat{z}_1 = \widehat{z}_2 - \widetilde{z}_2$.

We know $s(v;\mu) = \frac{1}{1+\exp(-v+\log 0.2-\log \mu)}$. From $\widetilde{z}_1 = 1 - \widetilde{z}_2$, we have:

$$\begin{aligned}
\frac{1}{1+\exp(-\widetilde{v}_2 + \log 0.2 - \log \mu)} &= 1 - \frac{1}{1+\exp(-\widetilde{v}_1 + \log 0.2 - \log \mu)} \\
\exp(-\widetilde{v}_1 - \widetilde{v}_2 + 2\log 0.2 - 2\log \mu) &= 1 \\
\widetilde{v}_2 &= 2(\log 0.2 - \log \mu) - \widetilde{v}_1 \qquad (40)
\end{aligned}$$

Then we have:

$$\begin{aligned}
\widehat{z}_1 + \widehat{z}_2 &= \frac{1}{1+\exp(-\widetilde{v}_1 - \varepsilon + \log 0.2 - \log \mu)} + \frac{1}{1+\exp(-\widetilde{v}_2 + \varepsilon + \log 0.2 - \log \mu)} \\
&= \frac{1}{1+\exp(-\widetilde{v}_1 - \varepsilon + \log 0.2 - \log \mu)} + \frac{1}{1+\exp(\widetilde{v}_1 + \varepsilon - \log 0.2 + \log \mu)} = 1
\end{aligned}$$

The last equation comes from the fact that $\frac{1}{1+\exp(x)} + \frac{1}{1+\exp(-x)} = 1$. Then we make use of the loss function expressed in terms of $\widetilde{z}$ and z, as defined in Eq. 31:

$$loss^{\text{econ}}_{\text{zz}}(\widetilde{z}, z; \mu) = \frac{\mu}{\widetilde{z}}(z - \widetilde{z})^2 + \mu(1 - \widetilde{z}) \qquad (41)$$

Since $\widetilde{z}_1 - \widehat{z}_1 = \widehat{z}_2 - \widetilde{z}_2$ and $\widetilde{z}_1 < \widetilde{z}_2$, that is $\frac{\mu}{\widetilde{z}_1} > \frac{\mu}{\widetilde{z}_2}$, we have the required result:

$$loss^{\text{econ}}_{\text{vv}}(\widetilde{v}_1, \widehat{v}_1; \mu) - loss^{\text{econ}}_{\text{vv}}(\widetilde{v}_1, \widetilde{v}_1; \mu) > loss^{\text{econ}}_{\text{vv}}(\widetilde{v}_2, \widehat{v}_2; \mu) - loss^{\text{econ}}_{\text{vv}}(\widetilde{v}_2, \widetilde{v}_2; \mu), \quad \forall \mu > 0. \qquad (42)$$

□

D VWAP Forecasts Across Size

Table D.6: VWAP forecasting accuracy by size groups (monthly VWAP)

model	jointly	nano	micro	small	large	mega
A. Equal-weighted relative absolute error (EW RAE) (OOS, %)						
ma_5	1.33	3.20	1.54	0.86	0.56	0.38
ols	1.13	2.70	1.32	0.74	0.45	0.33
nn	1.04	2.46	1.20	0.69	0.43	0.30
rnn	1.03	2.49	1.19	0.67	0.41	0.29
B. Volume-weighted relative absolute error (VW RAE) (OOS, %)						
ma_5	0.58	4.58	8.66	1.17	0.74	0.43
ols	0.48	3.70	4.12	1.75	0.58	0.37
nn	0.47	3.24	6.38	0.98	0.55	0.37
rnn	0.43	3.35	5.72	1.33	0.51	0.32

Note: This table expands Table 2 and reports average RAE by size groups. The first column "jointly" reports the overall RAE across all stocks, same as in Table 2.